



ENGINEERING CHEMISTRY

Lata Pasupulety

Department of Science and Humanities

ENGINEERING CHEMISTRY

Module 5- Energy storage devices



Module content:

- *Batteries*
 - *Components and working of a battery*
 - *Battery characteristics*
- *Modern batteries*
 - *Zinc Air battery*
 - *Lithium batteries*
 - *Li-ion battery (LiCoO_2)*
- *Reserve batteries*
- *Fuel cells*
 - *Principle , working and Efficiency of fuel cells*
 - *Types of Fuel cells*
 - *$\text{H}_2\text{-O}_2$ alkaline fuel cell*
 - *CO-O_2 solid oxide fuel cell*
 - *$\text{CH}_3\text{OH-O}_2$ polymer membrane fuel cell*
- *Oxygen sensors*
- *Supercapacitors*
- *Ragone plot*

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Module 5- Energy storage devices - Batteries



Class content:

- ***Batteries***
 - *Components of a battery*
 - *Working of a battery*
 - *Battery characteristics*

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Module 5- Energy Storage devices- Batteries



Battery

Device which stores chemical energy and converts it into electrical energy on demand

- **Alternate sources of energy**
- **Variable sizes and shapes**
- **Portable**
- Applications : Consumer goods - mobiles, laptops , calculators, flash lights
emergency lighting, space vehicles, electric vehicles, pacemakers

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Components of a battery

- **Anode** : Oxidation takes place

Electroactive material : metals which can undergo oxidation easily

Zn, Pb, Li

- **Cathode** : Reduction takes place

Electroactive material : compounds which can undergo reduction easily

PbO₂, MnO₂, O₂

- **Electrolyte**: Substance with good ionic conductivity

Acid, alkali or salt solutions; solids- doped oxides, polymers

H₂SO₄, KOH, Nafion

- **Separator**: Insulator which separates anode and cathode compartment

Role - To prevent internal short circuit

Transport ions from anode to cathode compartments and vice versa

Polypropylene, Cellophane

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Schematic diagram of a battery

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Module 5- Energy Storage devices- Batteries

Working of a battery

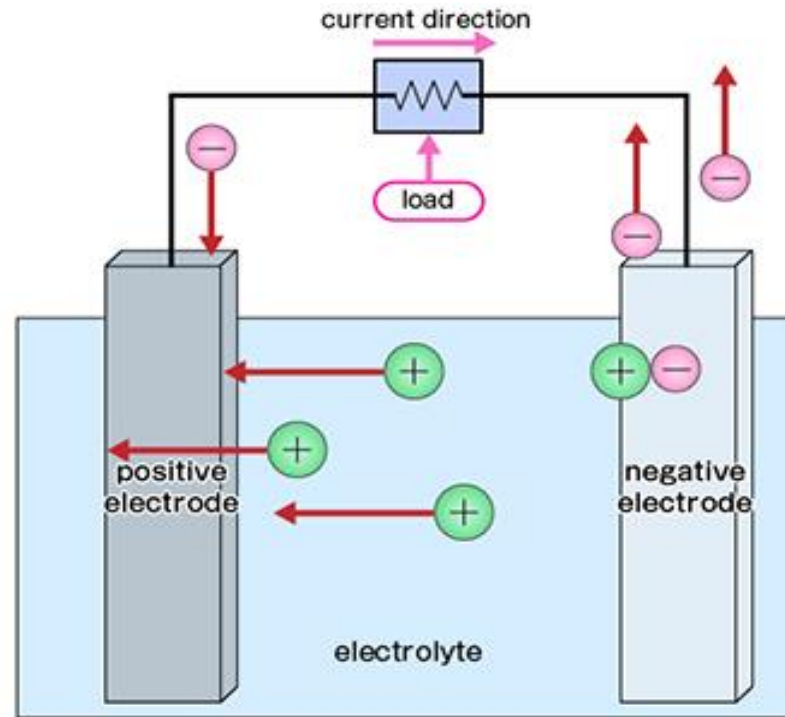
While discharging :

- Delivers power
- Galvanic cell

Anode: $M \rightarrow M^{n+} + ne^{-}$

Cathode: $A + ne^{-} \rightarrow A^{n-}$

Overall : $M + A \rightarrow MA$



<https://www.matsusada.com/column/secondary-battery.html>

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Module 5- Energy Storage devices- Batteries

Working of a battery

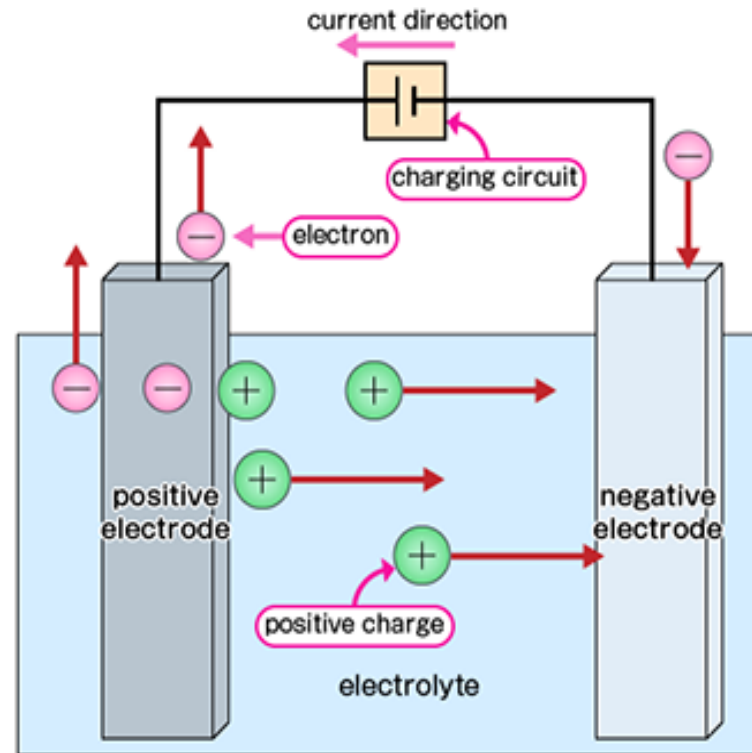
While charging:

- Consumes power
- Electrolytic cell

Anode: $A^{n-} \rightleftharpoons A + ne^{-}$

Cathode: $M^{n+} + ne^{-} \rightleftharpoons M$

Overall : $MA \rightarrow M + A$



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Based on working, two types of batteries:

Primary battery

- Electroactive material cannot be regenerated
- Discarded after electroactive species is consumed
- Galvanic cell
- Dry cell, Li-MnO_2

Secondary battery

- Electroactive material can be regenerated
- Can be used several times
- Galvanic as well as electrolytic cell
- Storage battery
- Li-ion, Pb-acid, Ni-Cd

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Battery characteristics

Voltage:

Emf of the cell $\rightarrow \Delta G \rightarrow$ reaction $\rightarrow$ choice of electrodes

$$E_{cell} = E^{\circ}_{cell} - \frac{2.303RT}{nF} \log Q$$

Factors affecting voltage of a battery:

- **E°_{cell}** ; higher the difference between $E^{\circ}_{cathode}$ and E°_{anode} , higher is the voltage of the cell
- **Temperature (T)** ; with increase in temperature voltage of the cell decreases
- **Reaction quotient (Q)** ; as the value of Q increases, the voltage of the cell changes marginally

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Current:

- Measure of the **rate** at which battery is discharging
- **Amperes, milliamperes**
- Depends on **rapid electron transfer reaction**
 - Amount of electroactive species
 - Conductivity of electrolyte
 - Inter-electrode distance



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Lata Pasupulety

Department of Science and Humanities

latapasupulety@pes.edu

+91 80 6666 3333 Extn 759



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Class content:

- ***Battery Characteristics(contd.)***
 - *Capacity*
 - *Energy storage density*
 - *Cycle life*
 - *Energy efficiency*
 - *Shelf life*
 - *Tolerance to service conditions*
 - *Power density*
 - *Energy density*

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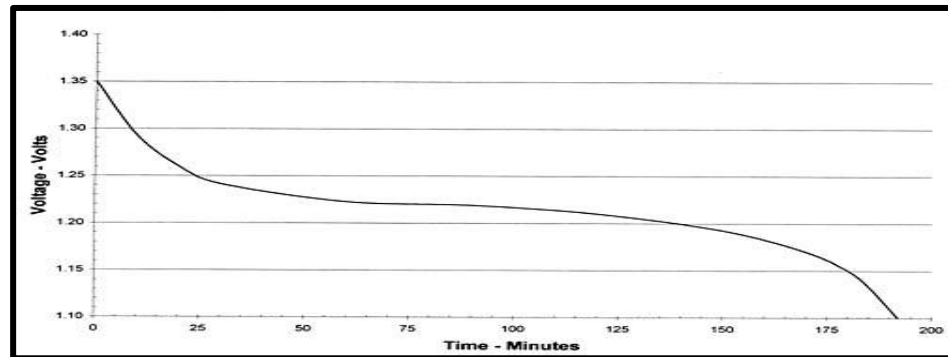
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Capacity:

- Charge or amount of electricity that may be obtained from the battery and is given in **Ah**
- Faraday's relation $C = \frac{wnF}{M}$ C is the capacity , w is the mass and M is the molar mass of active material
- Capacity depends on : **Size of the battery**
Discharge conditions of the battery

• $C = I t$

- The time period, t, for which the battery will last when a constant current, I, is drawn
- **Longer the flat portion** of the curve better is the capacity



http://www.hangtimes.com/robert_hoff_on_esv_s.html

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Electricity Storage density:

- Amount of charge or electricity per unit weight which the battery can hold
 - Weight includes weight of **all the components** of the battery - electrodes, electrolyte, case, current collectors, terminals etc
 - To get high electricity storage density, use of right kind of material for all the components is important
- e.g., **7g of Li** is required at anode to give **1F** of charge whereas **104g of Pb** is required for the same amount of charge

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Cycle life :

- Number of charge/discharge cycles that are possible before failure occurs
- Applicable only to **secondary batteries**
- **Reasons for limited cycle life:**
 - Corrosion at contact points
 - Shedding of active material from the plates
 - Shorting between electrodes due to irregular crystal growth and changes in morphology

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Energy efficiency:

$$\text{Percentage of Energy efficiency} = \frac{\text{Energy released on discharge}}{\text{Energy required for charging}} \times 100$$

- Depends on efficiency of electrode reactions

Shelf life:

- Maximum time for which a battery can be stored without loss of performance
- Low shelf life due to **self-discharge**

Tolerance to service conditions:

- The battery has to be tolerant to different service conditions such as variation in **temperature, vibration and shock**

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Energy density:

$$\text{Energy density} = \frac{\text{Energy available from the battery}}{\text{Mass of the battery}}$$

$$\text{Energy density} = \frac{i \times E_{\text{cell}}}{W} \times t$$

where W is mass of the battery, i and E_{cell} are current and EMF and t is time

- watt-hr/kg

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Module 5- Energy Storage devices- Batteries



Power density:

$$\text{Power density} = \frac{\text{Power available from the battery}}{\text{Mass of the battery}}$$

$$\text{Power density} = \frac{i \times E_{\text{cell}}}{W}$$

where W is mass of the battery, i and E_{cell} are current and EMF

- watt/kg



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Lata Pasupulety

Department of Science and Humanities

latapasupulety@pes.edu

+91 80 6666 3333 Extn 759



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Module 5- Energy storage devices - Batteries



Class content:

- *Modern batteries*
 - *Zinc – air battery*
 - *Construction*
 - *Working*
 - *Advantages*
 - *Disadvantages*
 - *Applications*

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Module 5- Energy Storage devices- Batteries



Modern batteries

Zinc air batteries

- Metal-air battery
- Anode : **Zn** ; Cathode : **O₂**
- Alkaline battery ; electrolyte : **alkali**
- Uses oxygen directly from the atmosphere to produce electrochemical energy
- Cathode active material need not be stored inside the battery
- **Energy density** is very high

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Module 5- Energy Storage devices- Batteries



Construction:

Anode : Zinc granules with gelling agent (to immobilize the composite and ensure adequate contact with Zinc granules) and a small amount of electrolyte

Cathode: Carbon (graphite) blended with MnO_2 (catalyst) with a wet proofing agent coated on nickel wire mesh support and an outer layer of air permeable Teflon layer. Air access holes on the cathode provide pathway for O_2 to enter the battery

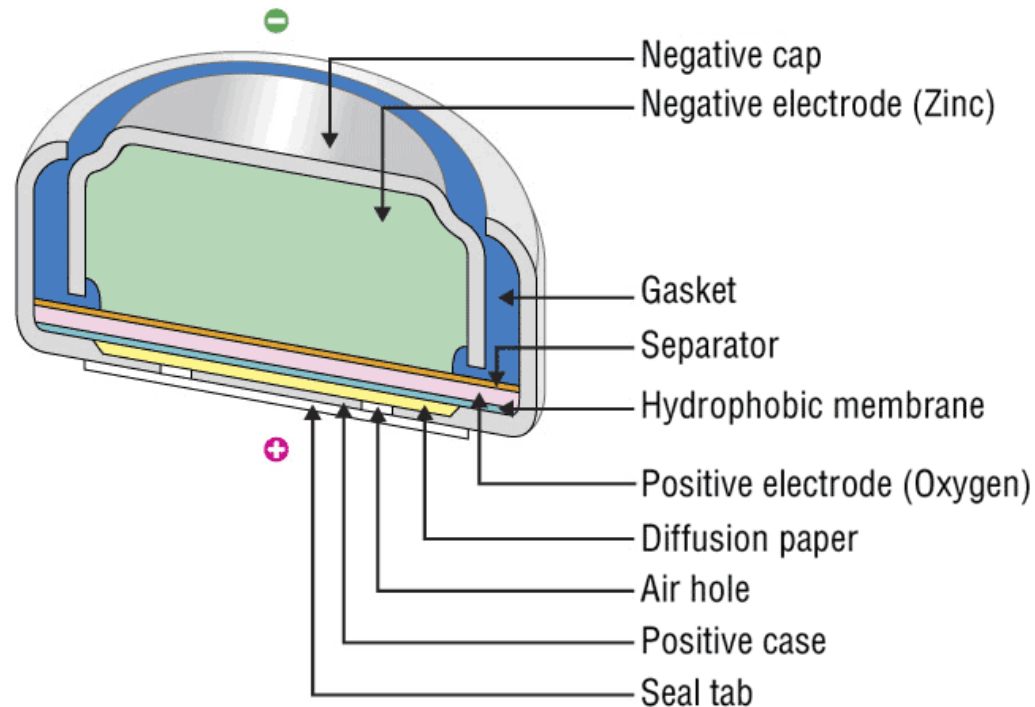
Electrolyte : 30% KOH

Separator : Polypropylene membrane soaked in electrolyte

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Cross-section of Zn-air battery

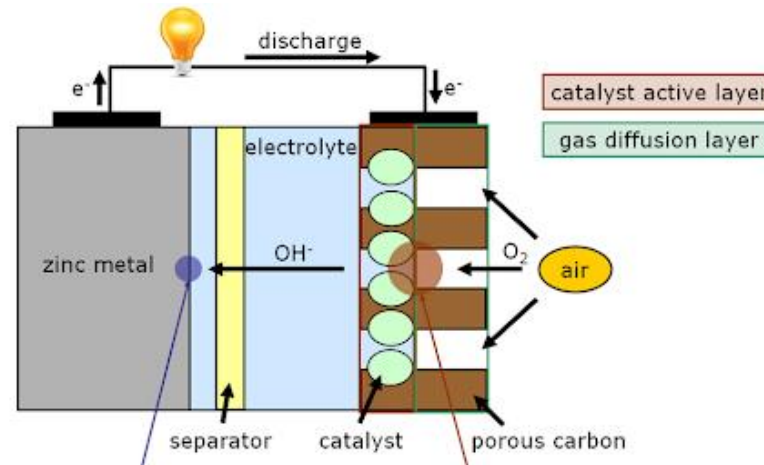
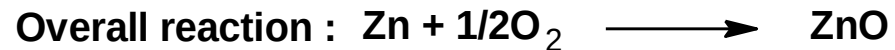
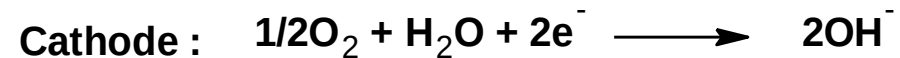
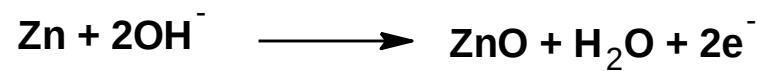
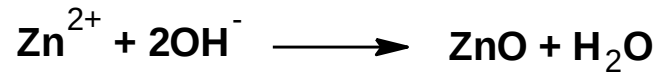
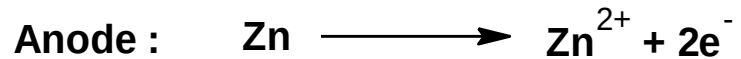


Source: <http://www.baj.or.jp/e/knowledge/structure.html>

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Working:



Source: <http://www.kosekgroup.cz/equipment/zinc-air-battery/>

Emf: 1.4 V

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Module 5- Energy Storage devices- Batteries



Advantages :

- High energy density : Air is taken directly from atmosphere and need not be stored ; doesn't contribute to the mass of the battery
- Very long shelf life : It can be kept sealed
- No ecological problems
- Low cost

Disadvantages :

- Limited power output
- Along with air, CO₂ may enter the battery. It reacts with KOH to form K₂CO₃, which will reduce the efficiency



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Applications :

- As power source in hearing aids
- In various medical devices
- In voice transmitters
- Large zinc-air batteries are used in rail-road signaling



THANK YOU

Lata Pasupulety

Department of Science and Humanities

latapasupulety@pes.edu

+91 80 6666 3333 Extn 759



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Module 5- Energy storage devices - Batteries



Class content:

- *Lithium batteries*
 - *Lithium - ion battery*
 - *Construction*
 - *Working*
 - *Advantages*
 - *Disadvantages*
 - *Applications*

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Module 5- Energy Storage devices- Batteries



Lithium batteries

- **Lithium is a popular anodic material:**
 - Light weight metal. The electrochemical equivalence of lithium is high (7g of lithium can give 1F of charge)
 - High negative standard reduction potential of -3.05V ;
when coupled with other electrodes gives high voltage
(about 4V)
- **Aqueous electrolytes cannot be used** as Lithium is very reactive and it reacts vigorously with water . So organic and inorganic electrolytes are used

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Components of Lithium batteries: Lithium is used as anode;

Cathode can be MnO_2 , SO_2Cl_2 . Electrolyte can be Li salt in organic solvents like acetonitrile, propylene carbonate or inorganic solvents like SOCl_2

Types of Lithium batteries :

Primary batteries which are not rechargeable, e.g. Li-MnO_2

Secondary batteries which are rechargeable, e.g. Li-ion battery

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Advantages of lithium batteries

- High Voltage up to 4V
- High energy density – Lightest metal
- High tolerance to service conditions (-40 °C to 70 °C)
- High electricity storage density
- Flat discharge characteristics

Disadvantages of lithium batteries

- Safety concerns due to high reactivity of Lithium metal
- Poor cycle life – due to dendrite formation
- Transportation limit

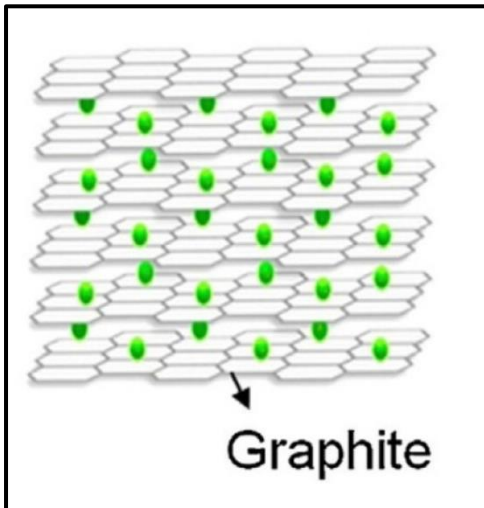
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Lithium ion battery

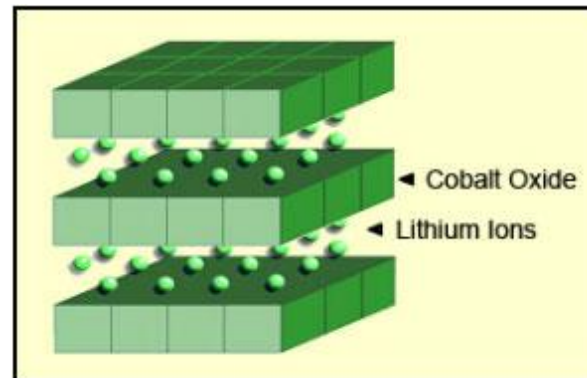
Principle :

- Lithium ion moves from anode to cathode while discharging and from cathode to anode while charging
- Materials used as anode and cathode should be capable of lodging Lithium ions
- Anode material: **Lithiated graphite**



Source:<https://www.jecst.org/journal/view.php?number=335&viewtype=pubreader>

Cathodic material : **LiCoO₂**



Source:https://batteryuniversity.com/learn/article/types_of_lithium_ion

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Lithium ion battery

Construction:

Anode: Lithiated -Carbon (Graphite) coated on Copper current collector

Cathode: Lithiated transition metal oxide coated on Aluminium current collector e.g. Lithium cobalt oxide(LiCoO_2)

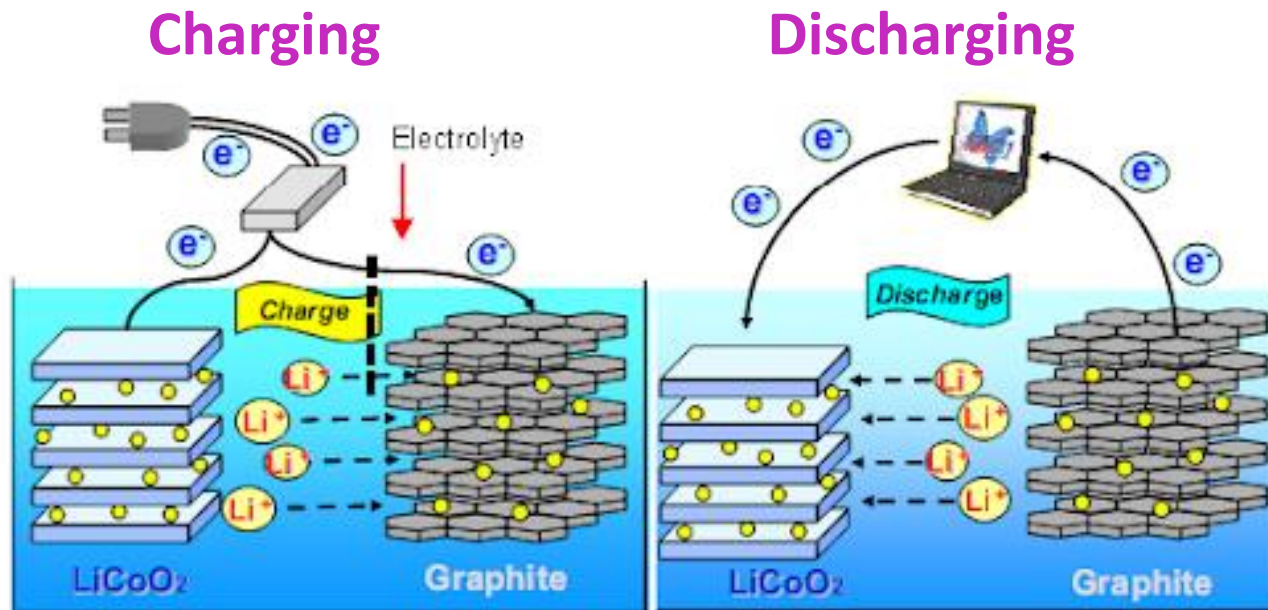
Electrolyte: The electrolyte is typically a mixture of organic carbonate solvents such as **ethylene carbonate or diethyl carbonate** containing **lithium salts like LiPF_6 , LiClO_4**

Separator: It is a very thin sheet of micro perforated polypropylene membrane

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Module 5- Energy Storage devices- Batteries

Working:



Source: <http://www.ee.ui.ac.id/epes/research-group/energy-materials>

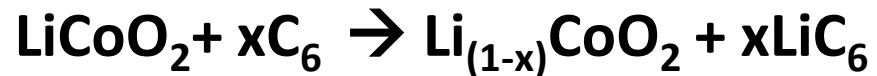
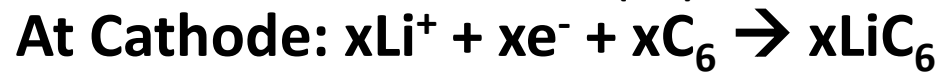
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Module 5- Energy Storage devices- Batteries

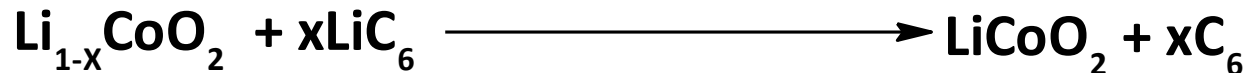
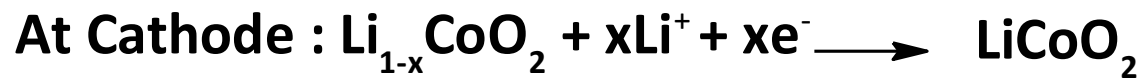
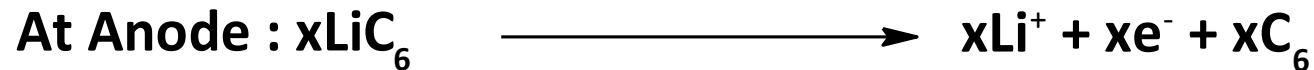


Reactions:

During Charging:



During Discharging :



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Advantages :

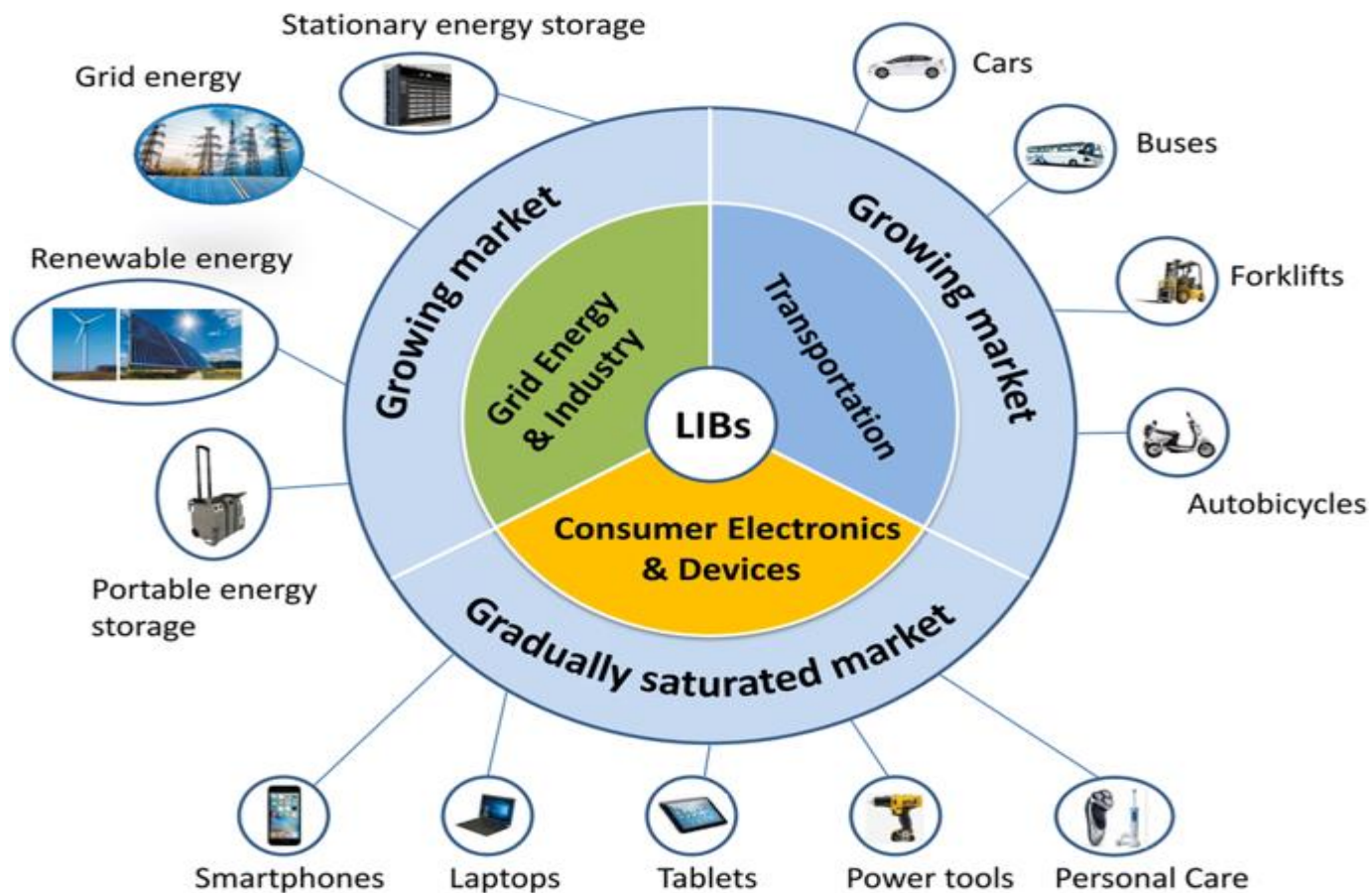
- Lighter than other rechargeable batteries for a given capacity
- Li-ion chemistry delivers a high open-circuit voltage 3.7 V
- Low self-discharge rate
- Do not suffer from battery memory effect
- Good cycle life as the problem of dendrite formation is eliminated(at no point, neither charging nor discharging, Lithium metal is formed)

Disadvantages :

- Rising internal resistance with cycling and age
- Safety concerns if overheated or overcharged

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Module 5- Energy Storage devices- Batteries



Source: Ding, Y. et al. *Electrochem. Energ. Rev.* **2**, 1–28 (2019).

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Module 5- Energy Storage devices- Batteries



Latest materials in use :

Lithium Nickel Cobalt Aluminum Oxide (LiNiCoAlO_2)

Lithium Nickel Manganese Cobalt Oxide (LiNiMnCoO_2) - Panasonic and Tesla

Lithium Titanate (Li_2TiO_3) - As anode; LiMnO_2 as cathode



THANK YOU

Lata Pasupulety

Department of Science and Humanities

latapasupulety@pes.edu

+91 80 6666 3333 Extn 759



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Lata Pasupulety

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Module 5- Energy storage devices – Fuel cells



Class content:

- *Fuel cells*
 - *Principle and working*
 - *Advantages*
 - *Disadvantages*
 - *Applications*

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Module 5- Energy Storage devices- Fuel cells

Fuel cells

Chemical energy → Thermal energy → Mechanical energy → Electrical energy



Fuel cell

- Galvanic cell which converts chemical energy of a fuel oxidant system **directly** into electrical energy by means of **redox reactions**
- **Do not store energy** but convert chemical energy in the fuel to electrical energy
- Fuel and oxidising agents have to be **continuously supplied** at the respective electrodes
- Generate power as long as the electroactive material is supplied

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Module 5- Energy Storage devices- Fuel cells



Construction:

- Two electrodes
- Catalyst
- Electrolyte
- Electroactive material supplied at the electrodes

Schematic diagram of a fuel cell

Fuel | Electrode | Electrolyte | Electrode | Oxidant

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Module 5- Energy Storage devices- Fuel cells



Working:

Anode : Fuel $\rightarrow$ Oxidation product + ne^-

Cathode : Oxidant + $ne^- \rightarrow$ Reduction products

Fuels : H_2 , CO, CH_3OH , C_2H_5OH , HCHO, N_2H_4

Oxidants: O_2 , H_2O_2 , halogens

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Module 5- Energy Storage devices- Fuel cells



Advantages :

- High power efficiency (about 50 - 80%)
- Ecofriendly - products formed are not toxic
- Silent operation

Disadvantages :

- Cost of the power is high as a result of the use of expensive electrodes and catalysts
- Power out put is moderate
- Fuels in the form of gases and oxygen need to be stored in tanks under high pressure-special equipment is required

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Module 5- Energy Storage devices- Fuel cells

Applications:

- Space exploration; as auxiliary power generators in space vehicles
- Vehicle traction for cars, buses and commercial vehicles
- Large scale power generation



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Module 5- Energy Storage devices- Fuel cells



Efficiency of a fuel cell:

- Thermodynamic efficiency is given by the ratio of **work output** to **heat input**
- Work output is **Gibb's free energy** ; heat input is **enthalpy change** of the cell reaction
- Gibb's free energy, ΔG - Measure of the electrical work done
- Enthalpy change, ΔH - Measure of the heat released during the reaction

$$\text{Efficiency \%} = \eta\% = \frac{\Delta G}{\Delta H} \times 100 = \frac{-nFE}{\Delta H} \times 100$$



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Lata Pasupulety

Department of Science and Humanities

latapasupulety@pes.edu

+91 80 6666 3333 Extn 759



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Module 5- Energy storage devices – Fuel cells



Class content:

- *Types of Fuel cells*

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Module 5- Energy Storage devices- Fuel cells



Types of Fuel cells (based on electrolyte):

- Alkaline fuel cell
- Phosphoric acid fuel cell
- Molten carbonate fuel cell
- Polymer electrolyte fuel cell
- Solid oxide fuel cell

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Module 5- Energy Storage devices- Fuel cells



Alkaline fuel cell

- Aqueous solution of **KOH** is used as electrolyte
- **Low temperature** fuel cell (operates at 100°C)
- Oxygen reduction is **more rapid** in alkaline electrolytes than in acid electrolytes
- Use of **non noble metal electro-catalyst** is feasible
- **Carbon containing fuels cannot be used** as CO₂ is formed as product which reacts with the electrolyte, KOH, to form K₂CO₃ which reduces efficiency of the cell



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Module 5- Energy Storage devices- Fuel cells



Phosphoric acid fuel cell

- Concentrated phosphoric acid is used as electrolyte
- Intermediate **temperature** fuel cell (operates between 160 – 220°C)
- Platinum is used as **electro-catalyst**
- Use **only H₂** as fuel
- H₂ used as fuel must be very pure as sulphur compounds and CO poison the Pt catalyst

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Molten carbonate fuel cell

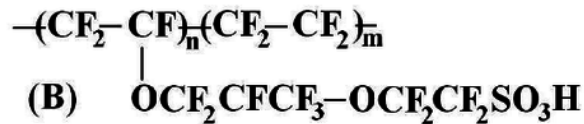
- Molten carbonates (mixture of $\text{LiAlO}_2 + \text{K}_2\text{CO}_3 + \text{Li}_2\text{CO}_3$) used as electrolyte
- High **temperature** fuel cell (operates between $600 - 650^\circ\text{C}$)
- Catalyst is not required since it operates at high temperature
- **H_2 or CO** can be used as fuel

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Module 5- Energy Storage devices- Fuel cells

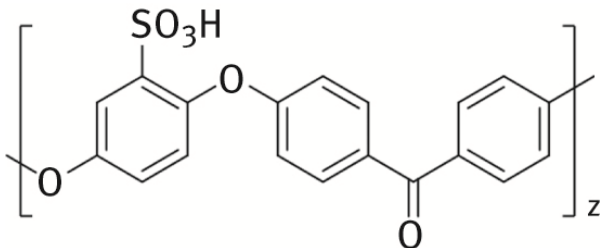
Polymer electrolyte membrane fuel cell

- Known as **proton exchange membrane fuel cell**
- **Polymer membrane** or proton exchange membrane is used as electrolyte
- Fluorocarbon backbone ($-\text{CF}_2-\text{CF}_2-$) similar to Teflon to which **sulphonic acid groups ($-\text{SO}_3\text{H}$)** are attached. The **protons on sulphonic acid group** are free to migrate through the hydrated membrane. e.g., (A) **Aquivion** and (B) **Nafion**



Source: https://www.researchgate.net/figure/Chemical-structures-of-A-AquivionR-or-Hyflon-and-B-NafionR_fig1_234842575

- New membranes are being used especially when CH_3OH is used as fuel - **Poly electrolyte membranes** e.g., **SPEEK** - sulphonated poly(ether ether ketone)



Source: <https://www.degruyter.com/view/journals/psr/2/8/article-20170018.xml?language=en>

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Module 5- Energy Storage devices- Fuel cells



Polymer electrolyte membrane fuel cell(contd.)

- **Low temperature fuel cell** (60- 90 °C)
- Polymer membrane **must remain hydrated** to maintain H^+ conductivity
- Water produced from the reaction must be removed from the cathode
 - **High temperatures** may dehydrate the polymer so H^+ conductivity cannot take place and the polymer may degrade and crack resulting in short circuit
 - **Low temperatures** will result in flooding of the cell thereby reducing efficiency of the cell and a higher catalyst loading will be required
- Low weight and volume
- **High energy density**
- **Noble metal catalyst** usually Pt is used
- **CO, if present poisons the catalyst** , so pure fuel and oxidant should be used

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Module 5- Energy Storage devices- Fuel cells



Solid oxide fuel cell

- **Ceramic oxide** capable of conducting oxide ions is used as electrolyte, e.g. ZrO_2 doped with Y_2O_3
- **Very high temperature** fuel cell (operates at $650 - 1000^\circ\text{C}$)
- Due to high temperature , **expensive catalyst need not be used**
- **CO** can also be used as a fuel
- **Slow** start - up



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Lata Pasupulety

Department of Science and Humanities

latapasupulety@pes.edu

+91 80 6666 3333 Extn 759



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Lata Pasupulety

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Module 5- Energy storage devices – Fuel cells



Class content:

- *$H_2 - O_2$ alkaline fuel cell*
 - *Principle*
 - *Construction and working*
 - *Advantages*
 - *Disadvantages*

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Module 5- Energy Storage devices- Fuel cells



Alkaline fuel cells

- Aqueous solution of **KOH** is used as electrolyte
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- Oxygen reduction is **more rapid** in alkaline electrolytes than in acid electrolytes
- Use of **non noble metal electro-catalyst** is feasible
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Module 5- Energy Storage devices- Fuel cells



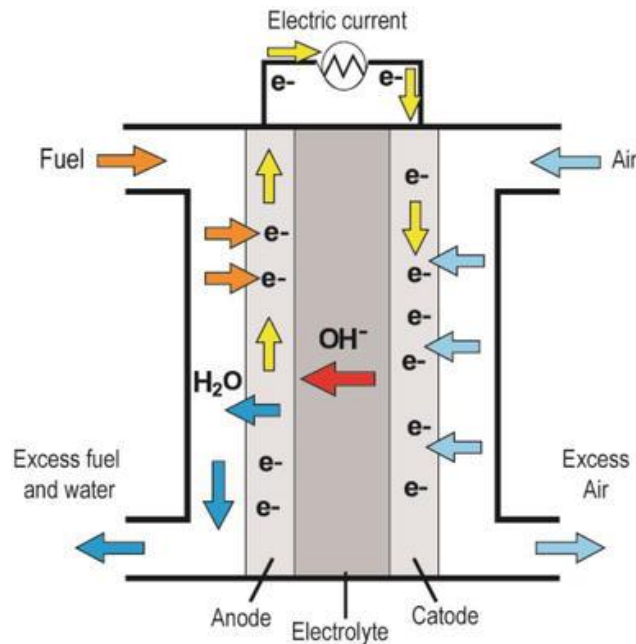
Construction :

- **Anode** : Porous carbon impregnated with Pt catalyst
- **Cathode** : Porous carbon impregnated with Ag catalyst
- **Fuel** : Hydrogen gas
- **Oxidant** : Oxygen gas
- **Electrolyte** : 30-45 % KOH(warm)

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Module 5- Energy Storage devices- Fuel cells

Cell representation:

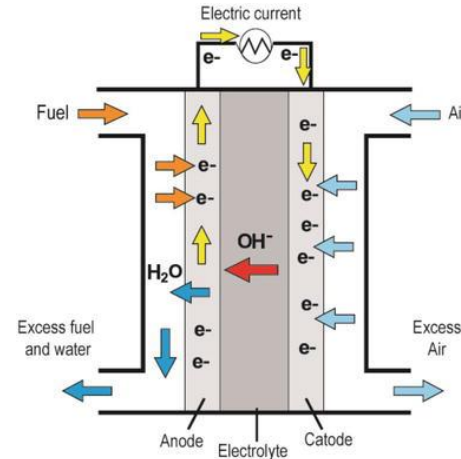
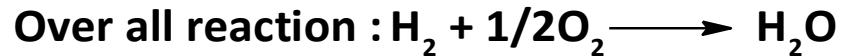
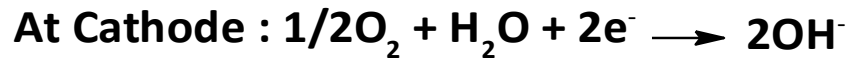
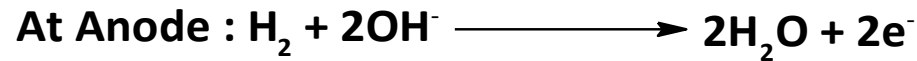


Source: <https://www.intechopen.com/books/new-trends-in-ion-exchange-studies/hydroxide-transport-in-anion-exchange-membranes-for-alkaline-fuel-cells>

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Module 5- Energy Storage devices- Fuel cells

Working:



- H_2 gas diffuses through anode, gets adsorbed on the electrode surface, reacts with OH^- to form water
- At cathode O_2 diffuses through electrode, is adsorbed and reduced to OH^-
- Product is water which dilutes the KOH
- Cell operates at 100°C , so that water from KOH escapes as steam
- The water was used by astronauts for drinking on Apollo spacecraft

Emf = 1.23 V

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Module 5- Energy Storage devices- Fuel cells



Advantages:

- Operates at low temperature
- Alkali is used as electrolyte hence non noble metal catalyst can be used so less expensive

Disadvantages:

- Reactants must be free from C, because on oxidation CO_2 is formed. The alkali reacts with CO_2 to form carbonates which reduce efficiency of the cells; pure fuel and oxidant which are free of carbon compounds must be used
- Liquid electrolytes pose handling problems



THANK YOU

Lata Pasupulety

Department of Science and Humanities

latapasupulety@pes.edu

+91 80 6666 3333 Extn 759



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Lata Pasupulety

Department of Science and Humanities

ENGINEERING CHEMISTRY

Module 5- Energy storage devices – Supercapacitors



Class content:

- ***Supercapacitors***
 - *Principle*
 - *Construction and working*
 - *Advantages*
 - *Disadvantages*
 - *Applications*
- ***Ragone Plot***

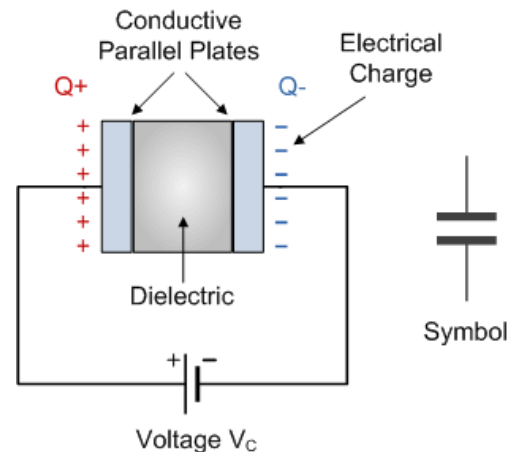
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Module 5- Energy Storage devices- Supercapacitors

Capacitor

- A two- terminal electrical component that has the **ability to store energy** in the form of electrical charge
- Has two conducting plates separated by a dielectric
- When a DC voltage is connected across the capacitor, one plate becomes positive and the other negative
- The **charge accumulation on the plates** causes a voltage or potential difference across the capacitor

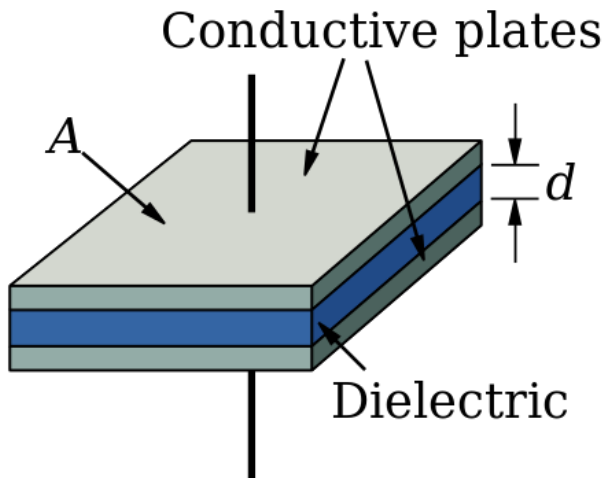
Source:https://www.electronics-tutorials.ws/capacitor/cap_1.html



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Module 5- Energy Storage devices- Supercapacitors

- **Capacitance** is the charge accumulation capability of a capacitor
- Charge (Q) stored in a capacitor is directly proportional to the voltage (V) developed across the capacitor; $Q \propto V$
- $Q = C \times V$, where C is the capacitance measured in **farads**
- Capacitance is directly proportional to **area of plates (A)** and inversely proportional to the **distance between the two plates (D)**; $C \propto \frac{A}{d}$



Source: https://www.wikiwand.com/en/Capacitor_typeS

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Module 5- Energy Storage devices- Supercapacitors



Supercapacitors

- Also known as **ultra capacitors** or **electrochemical double-layer capacitors(EDLC)**
- Are energy storage devices that have **high capacitance** and are used to store large amounts of electrical charge
- **Charge and discharge** very quickly
- Capacitance is several thousand times that of a Capacitor
 - **Capacitor ----- μF**
 - **Supercapacitor ----- thousands of Farads**

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Module 5- Energy Storage devices- Supercapacitors



Construction:

Electrodes: made of high surface area materials such as porous carbon, graphene, carbon nanotubes and certain conducting polymers or carbon aerogel

Electrolyte: KOH, H_2SO_4 , Na_2SO_4

Separator: an ion permeable separator is placed between the electrodes in order prevent electrical contact, but still allows ions from electrolyte to pass through e.g., porous polypropylene

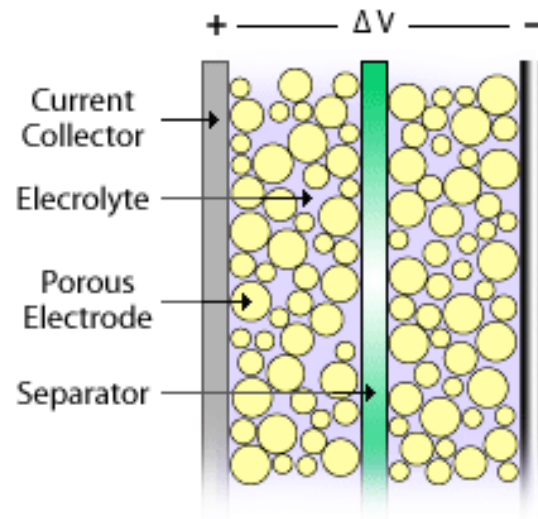
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Separator is sandwiched between the electrodes

This is placed into a case , impregnated with **electrolyte** and sealed

The electrodes are flanked with **current collectors**



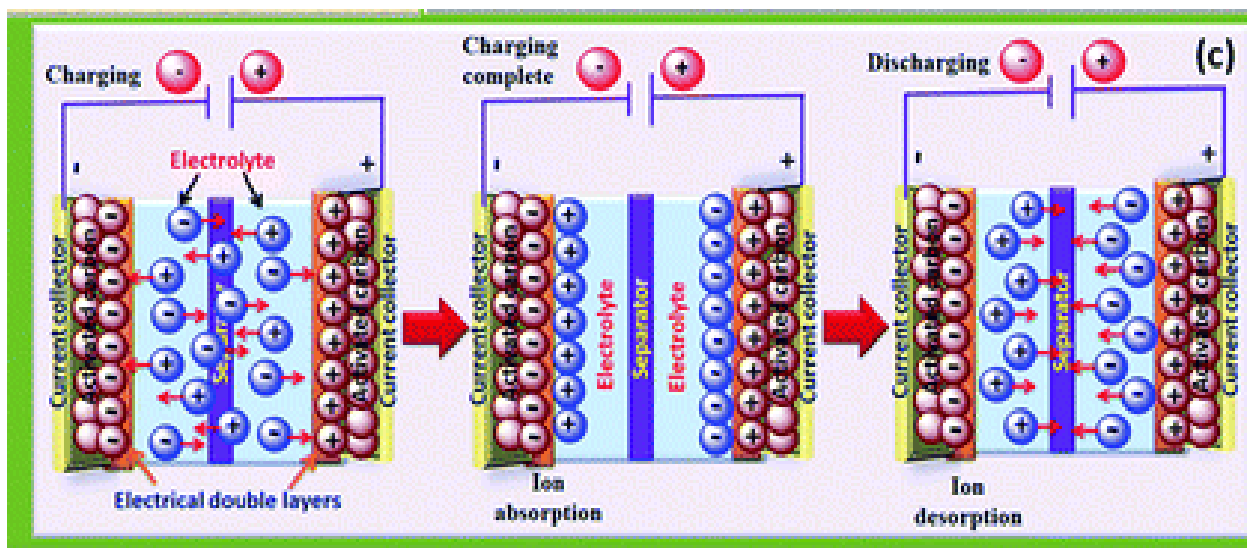
Source: <https://www.azonano.com/article.aspx?ArticleID=3044>

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Module 5- Energy Storage devices- Supercapacitors

Working:

- When a potential is applied, the positive electrode attracts negative ions in the electrolyte, while negative electrode attracts the positive ions
- Formation of electrical double layer at entire electrode /electrolyte interface with a charge separation in nanometer scale



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Module 5- Energy Storage devices- Supercapacitors



- Only **absorption and desorption of ions** takes place at the electrode during charging and discharging and there are **no redox reactions**
- Distance between the charged layers in nanorange, **d is very small** and use of porous electrodes gives very high surface area , **A is large**

$$C \propto \frac{A}{d} ; \text{Capacitance is very high}$$

- Formation of the electrical double-layer at each electrode/electrolyte interface, overall capacitance (C_T) is expressed as:
$$\frac{1}{C_T} = \frac{1}{C_+} + \frac{1}{C_-}$$

where C_+ is the capacitance of cathode/electrolyte interface and C_- is the capacitance of anode/electrolyte interface

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Module 5- Energy Storage devices- Supercapacitors



Advantages:

- **Rapid charging**; charge in a few seconds
- **High power density** as they discharge very fast
- **High cycle life**, can be cycled millions of time
- **Safe** as extremely low internal resistance and extremely low heating rates

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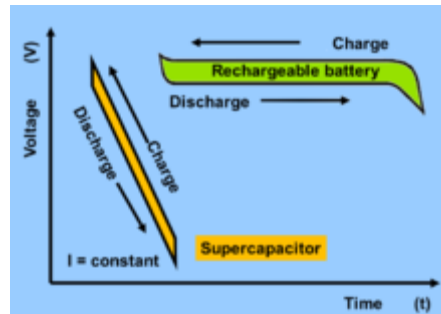
Module 5- Energy Storage devices- Supercapacitors

Disadvantages:

Low energy density

High self discharge

Linear discharge voltage



Source: <https://store.chipkin.com/articles/beyond-the-lithium-ion-battery-a-look-at-supercapacitors-and-other-batteries>

High cost

Power available for a short duration

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Module 5- Energy Storage devices- Supercapacitors



Applications:

- Memory back-up
- Hybrid cars for start-stop application
- Flash photography devices in digital cameras, flash lights, portable media players
- As an intermediate energy storage for FM radios, cell phones, and emergency kits

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Module 5- Energy Storage devices- Supercapacitors

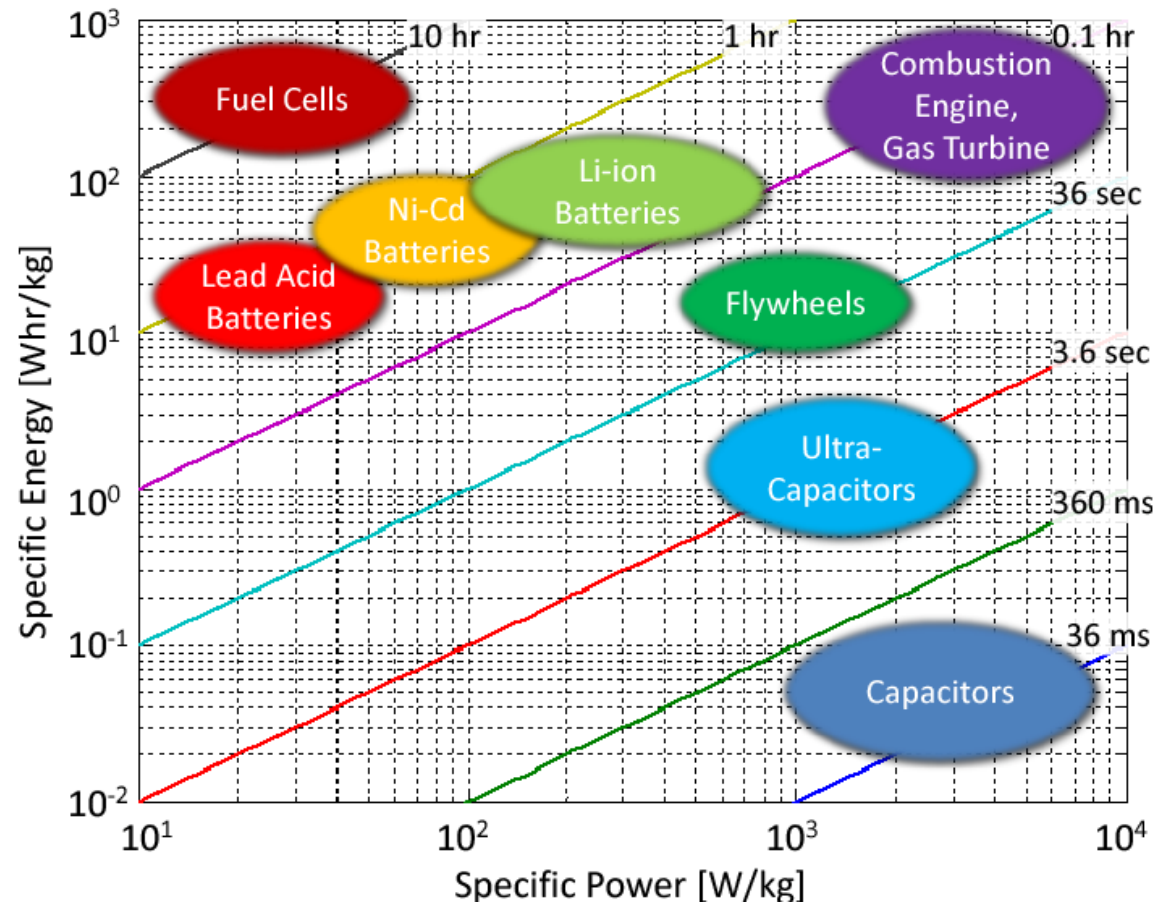


Ragone Plot

- **Energy density** (Wh/kg) is plotted against **Power density** (W/kg)
- To **compare performance** of various energy storage devices
- Since it uses a double-logarithmic chart, storage technologies with very different storage properties can be compared in one plot

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Module 5- Energy Storage devices- Supercapacitors



Source: <https://www.semanticscholar.org/paper/Education-on-vehicle-electrification%3A-Battery-Fuel-Moura-Siegel/7e83f03396b55f2a894461480dda7e9c154f5721/figure/0>

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Module 5- Energy Storage devices- Supercapacitors



- **Fuel cells** have high energy density as the electroactive species can be continuously supplied but the power density is low due to slow kinetics of redox reactions at electrodes
- **Battery systems** offer moderate energy density and power density.
- **Ultracapacitors** (supercapacitors) can deliver very high power density as they can discharge a large amount of charge quickly (because no redox reaction is involved) but energy density is very limited because the charge cannot be stored for a long time

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Module 5- Energy Storage devices- Supercapacitors



End of module 5

End of Unit III



THANK YOU

Lata Pasupulety

Department of Science and Humanities

latapasupulety@pes.edu

+91 80 6666 3333 Extn 759



ENGINEERING CHEMISTRY

Dr. M. Veerabhadraswamy

Department of Science & Humanities,
Mobile no. 9241503423 e-mail: vbs@pes.edu

Module content:

- *Hydrogen Energy*
- *Hydrogen Production methods*
 - *Steam reforming*
 - *Electrolysis of water*
- *Hydrogen Storage*
 - *Methods of Storage*

Class content:

- *Hydrogen Energy*
- *Hydrogen Production methods*
 - *Steam Reforming*
 - *Electrolysis of water*

Hydrogen Energy

- The Hydrogen energy has been identified as one of potential sources of fuel in the state-of-art technological era
- Hydrogen production, storage and transportation has become the focus of science and technology
- Sustainable production of Hydrogen from renewable feedstocks and electrolysis of water has taken lead role in technology development
- More emphasis is provided to understanding and engineering the production and storage of Hydrogen produced by Electrolysis method

Advantages

- Abundant availability in earth crust
- Compatibility of hydrogen with fuel cell
- High efficiency (65%) than diesel (45%) and gasoline (22%)

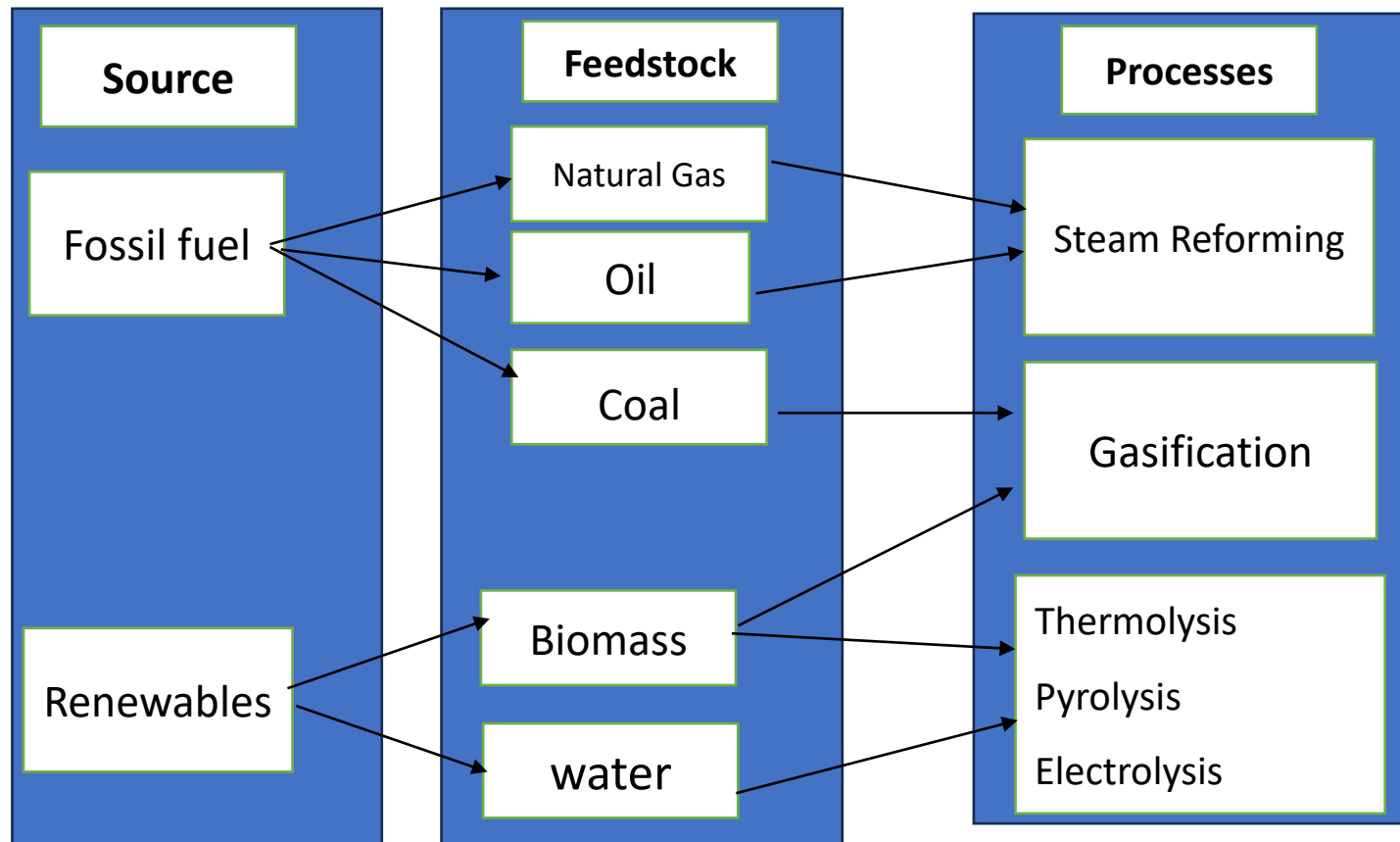
Disadvantages

- High cost
- Highly flammable
- It is still dependent on fossil fuels

Hydrogen Economy

- Hydrogen used as fuel reduces pollution
- It is carbon free energy source
- **Hydrogen economy** is a vision of an energy delivery infrastructure based on hydrogen as a carbon-free energy carrier
- It provides
 - alternative energy economy in the form of a parallel power and transportation infrastructure
 - technical solutions for energy-efficient hydrogen production, storage technology and delivery infrastructure

Strategic Hydrogen production methods



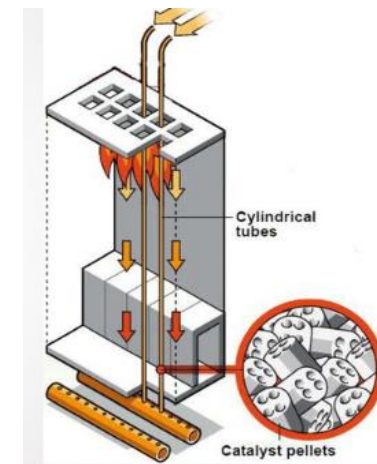
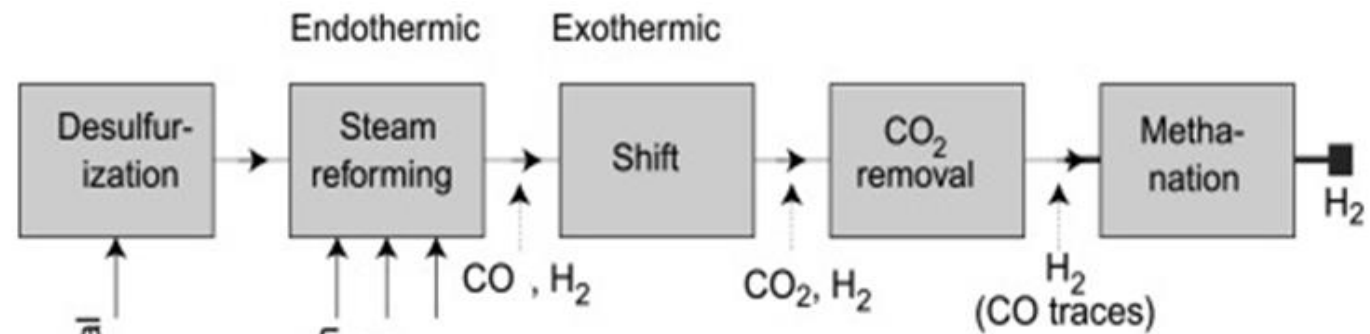
Hydrogen Production methods

- Steam reforming (steam methane reforming –Grey Hydrogen)
 - Natural gas is used to produce Hydrogen
 - Most widespread process for generation of hydrogen
- Gasification process (Black Hydrogen)
 - Hydrogen is produced from coal by gasification process
 - Coal is converted into CO_2 and subjected to gasification process to produce Carbon monoxide which is subjected to water gas shift process to produce mixture of CO_2 and H_2 is purified to get pure H_2
- Thermolysis (Green Hydrogen)
 - involves thermochemical water splitting at high temperature 500°C – 2000°C to produce hydrogen
 - carried in a closed system-the chemical used in the process are reused within each cycle consumed only water and produces hydrogen and oxygen

- Photolysis
 - Processes use light energy to split water into hydrogen and oxygen
 - One of the potential for sustainable hydrogen production
 - Low environmental impact
 - Yet in early stage of research
- Electrolysis
 - uses electricity to split water into hydrogen and oxygen
 - Electrolysis reaction takes place in a unit called an **electrolyzer**
 - Method is promising option for carbon-free hydrogen production from renewable and nuclear resources.

Steam reforming (Steam methane reforming)

- Process involves four important steps
- Desulphurization
 - removal of sulphur from the feedstocks
 - Sulphur tends to poison catalysts used in the steam reforming
- Steam reforming
 - Steam, fuel and air in the reform chamber in presence of catalyst form syngas
 - $\text{CH}_4 + \text{H}_2\text{O} \rightarrow \text{CO} + 3 \text{H}_2$
 - 850-900°C
 - Ni catalyst
 - Endothermic reaction ($\Delta H^\circ = +206 \text{ KJ/mole}$)
 - High temperature and low pressure is favoured



- Shift reforming (Water Gas shift reactions)



- Catalyst: $\text{Fe}_3\text{O}_4/\text{Cr}_2\text{O}_3$
 - Exothermic reaction ($\Delta H^\circ = -41.0 \text{ KJ/mole}$)
 - Favoured at low temperature (350°C)
 - H_2 produced will have traces of Carbon monoxide as impurities
- Purification
 - CO_2 is removed to product pure Hydrogen
 - Traces of CO is removed by **methanation** ($350 - 450^\circ\text{C}$)
 - $\text{CO} + 3\text{H}_2\text{O} \rightarrow \text{CH}_4 + \text{H}_2\text{O}$

Advantages and Limitations of Steam reforming method

Advantages

- General produces higher yield H_2 (50%)
- Heat generated can be recycled to increase efficiency
- Relatively stable during process

Limitations

- High level of carbonaceous materials are formed
- External heat source is needed to initiate the reaction

Green Hydrogen production

Green Hydrogen refers to production of Hydrogen gas from renewable resources

Green Hydrogen production will help us reach both the momentum and promise for greener mobility and industry

Hydrogen production through Electrolysis produces clean energy with pollution free process by using electricity

Hydrogen production by electrolysis can be achieved by three methods.

1. Alkaline Electrolysis
2. Proton Membrane Electrolysis
3. High temperature Electrolysis

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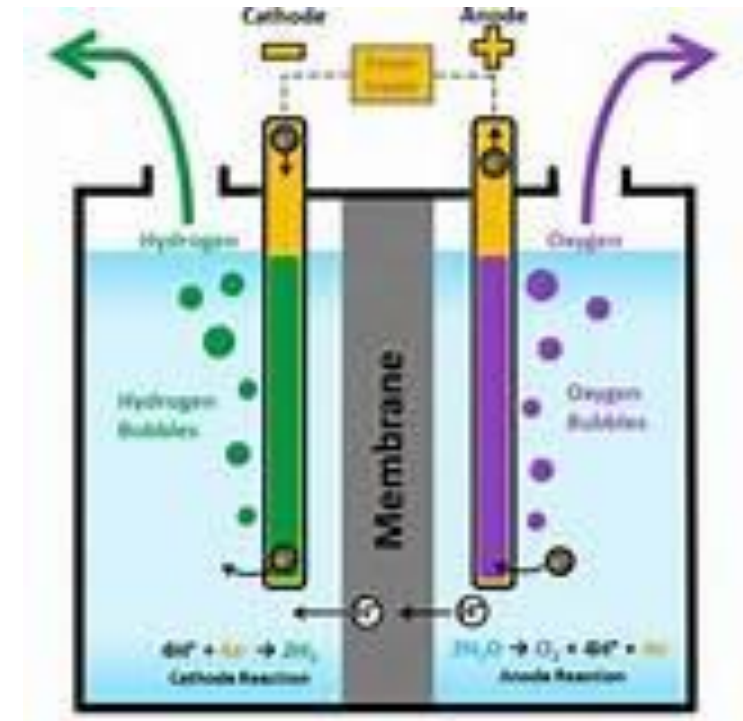
Module 6- Corrosion Chemistry

Working principles of Electrolysis of water:

- When water is subjected to electrochemical process water molecules split in hydrogen and oxygen
- Electricity is used for splitting the hydrogen and oxygen into their gaseous phase
- Alkaline Electrolysis: Alkaline Process
- The Hydrogen production by alkaline water electrolysis is one of the environmentally friendly



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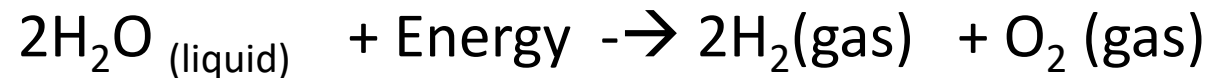


Electrolysis of water

<https://th.bing.com/th/id/OIP.y4cn6pR0TZFTSuf22XtXLgAA?pid=ImgDet&w=161&h=159&c=7>

- Zero emission of CO_2 (if this process combined with renewable energy sources like electricity from solar or wind energy)
- Alkaline water electrolysis is conventional methods with easiest, simplest, and suitable methods for Hydrogen production
- Electrolysis requires needs Direct current supply
- Electrolyser split water into Hydrogen and Oxygen gas

- Splitting of water is very small approximately 7- 10 moles/ litre
 - because pure water is the very poor conductor of electricity hence acid (H_2SO_4 in acid electrolyzer) or base (KOH or NaOH in Alkaline electrolyzer) is used to improve the conductivity
- Solutions split into ions which improving conductivity of the electrolyte.
- The Reaction of water splitting is as follows:
 - At Anode: $2\text{OH}^- \rightarrow \frac{1}{2} \text{O}_2 + \text{H}_2\text{O} + 2 \text{e}^-$
 - At Cathode: $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$
- The basic equation of water splitting is represented as:



- In Alkaline electrolyte process
 - aqueous solution containing either NaOH or KOH used
 - typical concentration of 20-40 wt %
 - operation temperatures are between 343 – 363 K
 - operating pressure up to 3 MPa
- Charge carrier: OH^-
- Anodic materials: Ni and its alloys
- Cathodic materials: Cd, Pb, Cu, Ag, Pt, Pd.
- Electrolyte materials : KOH, NaOH
- Catalyst materials: Ni-Zn, Ni-Co-Au, RuO_2 , LaCoO_3

Advantages and Limitations of Electrolysis process

Advantages

- No harmful emission produced
- Clean hydrogen production
- Results in zero greenhouse gas emission
- Used in Fuel cells

Limitations

- Cost of production is high
- Efficiency is low
- Need high power source
- Storage is challenge

- **Conclusion:**
- The Hydrogen energy has been identified as one of potential sources of fuel, Production of Green Hydrogen is the key focus to meet the sustainable development.
- Hydrogen is not available as a pure gas, it must be produced from other compounds like fossil fuels, biomass, electrolysis, etc.,
- Electrolysis of water to produce hydrogen is one of the most viable technology
- Hydrogen evolution takes place at cathode materials, whereas oxygen evolves takes place at anode materials.
- Catalysts improve the performance of the electrode materials to generate high and pure hydrogen gas.



THANK YOU

For any query contact: VBS

Dr. M. Veerabhadraswamy

Department of Science & Humanities,
e-mail: vbs@pes.edu, vsgswamy@yahoo.co.in
Mobile: +91 9241503423, 8310425312

<https://scholar.google.co.in/citations?user=fkjjyv4AAAAJ&hl=en>



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Dr. M. Veerabhadraswamy
Department of Science & Humanities,
Mobile no. 9241503423 e-mail: vbs@pes.edu

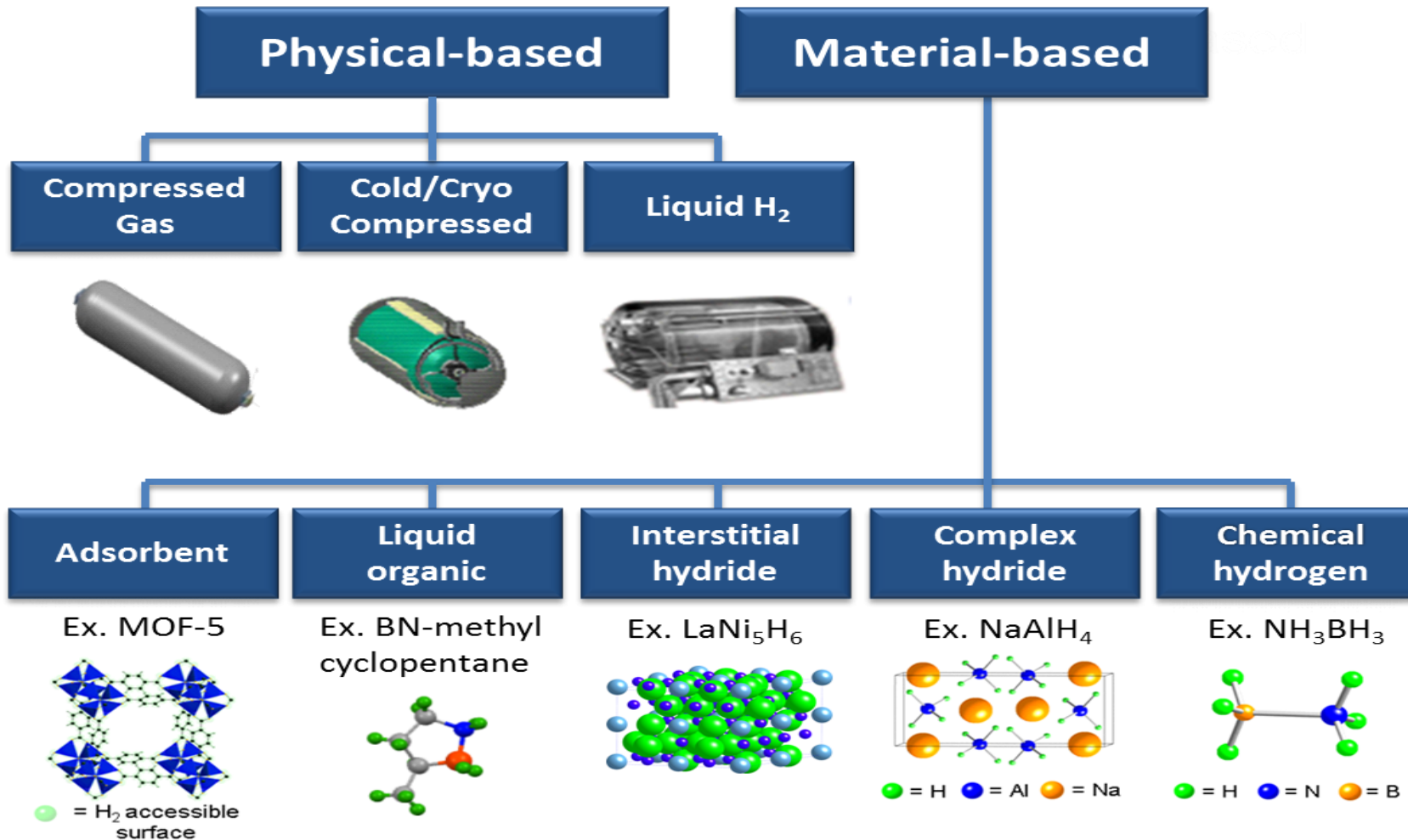
Module content:

- *Hydrogen Storage*
 - *Methods of Storage*

Hydrogen Energy

- The Hydrogen storage is crucial for effective utilization of Hydrogen for its versatile applications.
- Methods used for hydrogen storage:
 - Liquid form
 - Gaseous
 - Solid State

How is hydrogen stored?



- [Hydrogen Storage in Metal-Organic Frameworks: A Review - research journal \(gyanvihar.org\)](#)

- Hydrogen can be physically stored as either a gas or a liquid
- Storage of hydrogen in the form of gas usually requires high-pressure tanks (350–700 bar tank pressure)
- Storage of hydrogen as a liquid requires cryogenic temperatures because the boiling point of hydrogen is -252.8°C at an atmospheric pressure
- Hydrogen can also be stored on the surfaces of solids (adsorption) or within solids (absorption)

Solid stage hydrogen storage:

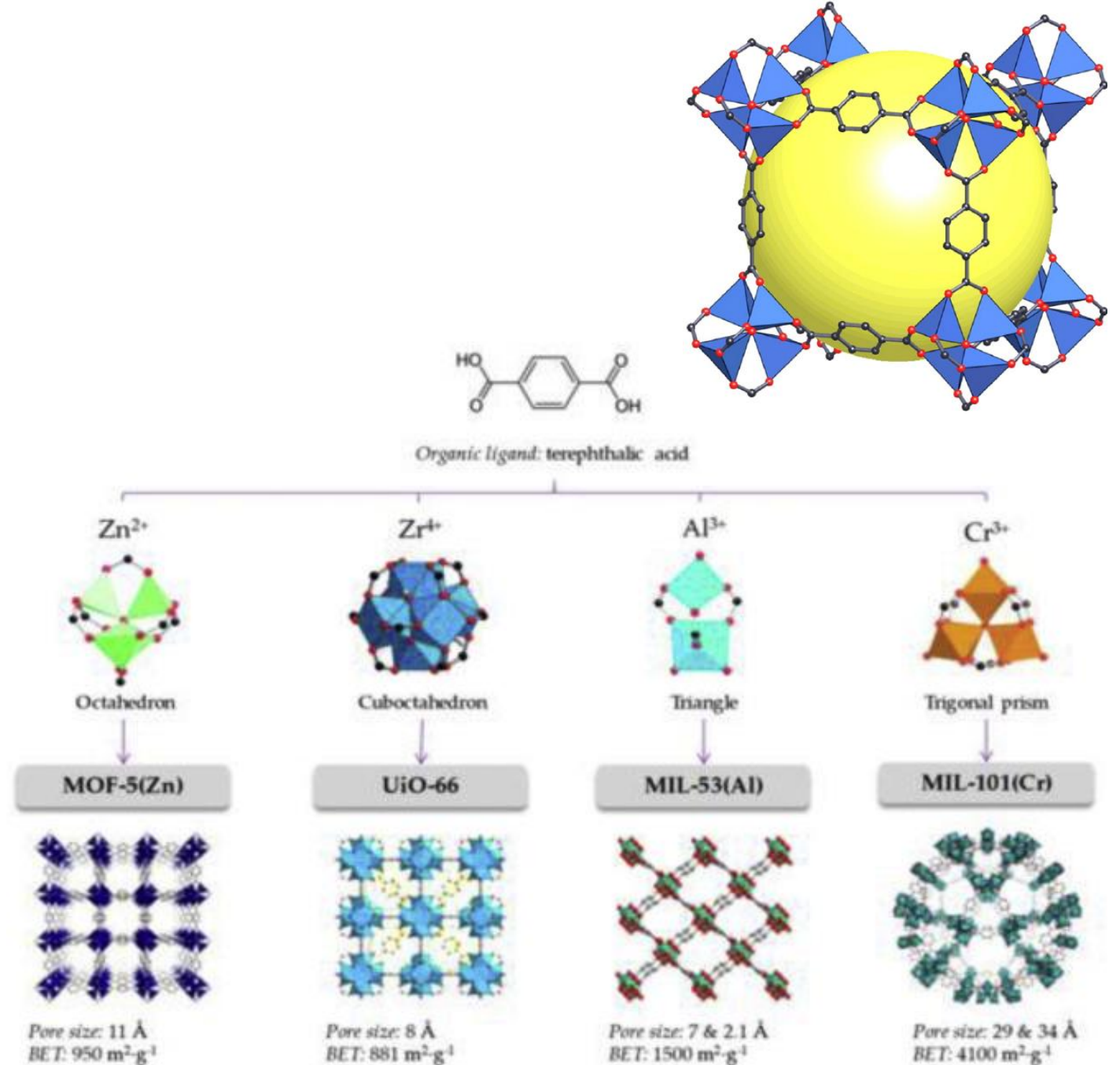
- Depending on the type of hydrogen being stored in host materials, solid-state hydrogen storage can be classified into two categories:
 - Chemisorption of hydrogen atoms
 - Physisorption of intact hydrogen molecules
- In chemisorption (absorption into matter), binding of hydrogen atoms occur into the surface
- In case of physisorption (adsorption on surface)
- The Hydrogen molecule is bound onto the surface of adsorbent

Some of the most prominent materials for Hydrogen Storage includes

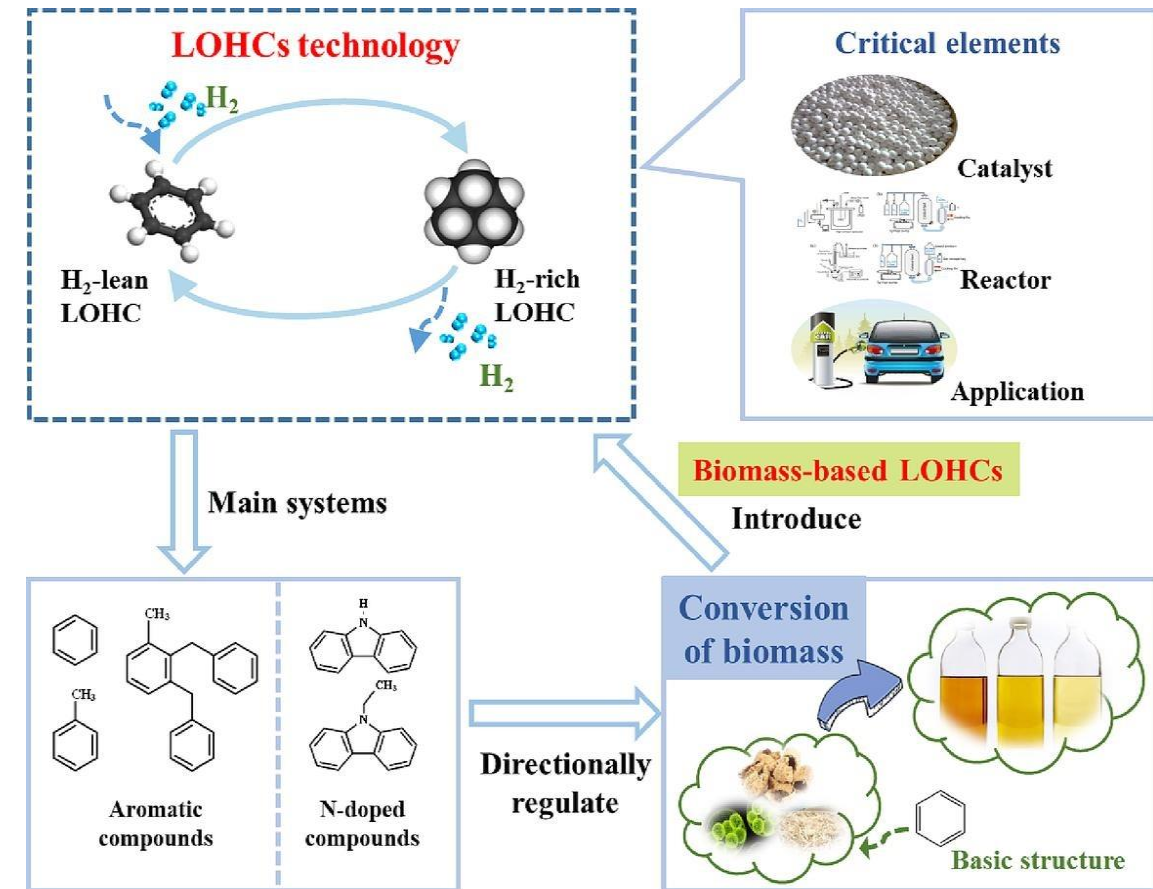
- Metal organic framework (MOF) as absorbent
- Liquid organic hydrogen carriers (LOHC)
- Interstitial hydride
- Complex hydride
- Chemical Hydrogen storage

- Metal Organic Frame as Absorbent: (MOF)

- representing a novel class of porous materials
- feature unique pore structure, such as exceptional porosity, tunable pore structures
- Enables high density energy storage of clean fuel gas in MOF adsorbents
- Zn-MOF (MOF-5) remains an attractive hydrogen storage material due to its balanced gravimetric and volumetric hydrogen uptake



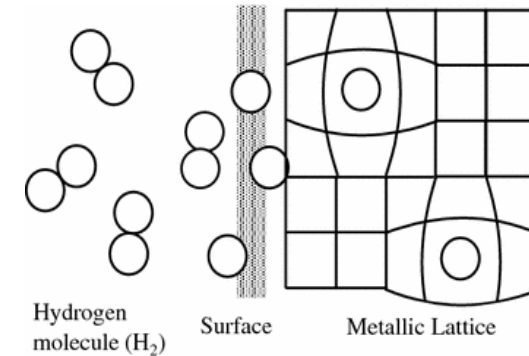
- Liquid organic hydrogen carriers (LOHC)
 - LOHC technology has shown great potential for efficient and stable hydrogen storage and transport
 - This technology allows for safe and economical large-scale transoceanic transportation and long-cycle hydrogen storage
 - LOHCs can be used as storage media for hydrogen
- Organic compounds that can absorb and release hydrogen through chemical reactions
 - Methyl cyclopentane
 - Dibenzyl toluene



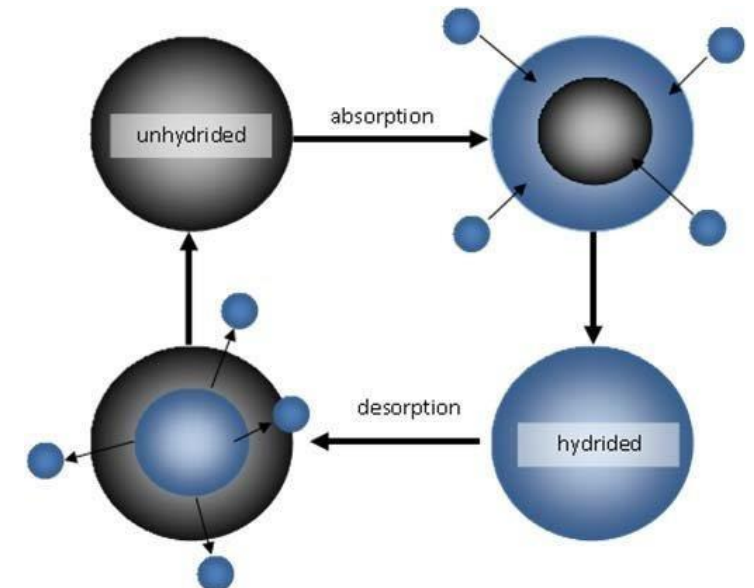


Interstitial hydride

- Contain a larger quantity of hydrogen than the same volume of liquid hydrogen
- Have advantage in the amount of hydrogen on a weight basis
- Certain interstitial hydrides are very suitable for hydrogen storage and transportation
- Example: LiNi_5H_6

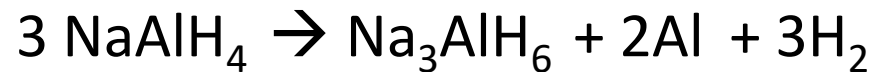


[Solid Hydrogen Storage Materials: Interstitial Hydrides | SpringerLink](#)



- Complex hydrides
 - Composed of metal cations (such as Li, Mg, Na, etc.) and hydrogen-containing coordination anions (such as AlH_4^- , NH_2^- , BH_4^-)
 - On heating, the metallic hydride decomposes to hydrogen and finely divided metal

Example: Sodium aluminate can release up to 7.4% of Hydrogen when heated at 200°C



- Chemical Hydrogen
- Hydrogen is released from a material through a chemical reaction
- Hydrogen is restored through a chemical reaction when the material is being recharged
- Ammonia Borane ($\text{NH}_3\text{-BH}_3$) has exceptional properties for chemical hydrogen gas storage
- Hydrolysis of $\text{NH}_3\text{-BH}_3$ releases hydrogen stored in it in presence of catalyst

Advantages

Disadvantages

- Liquid form

- Liquid H₂ exist at low temperature 20K (-253K)
- Stored in cryogenic tanks
- Low volume compared gaseous H₂

- Reinforced & insulated storage tank required
- Cooling and compression process consume energy (net loss about 30%)

- Gaseous form

- It involves less infrastructure
- More cost effective
- Can be compressed into high pressure in gas form

- Needs extra energy
- Space occupied is large
- Leak proof tanks needed

- Solid form

- hydrogen can be released whenever required by thermal stimulation or any other technique
- Ease of handling
- Safety

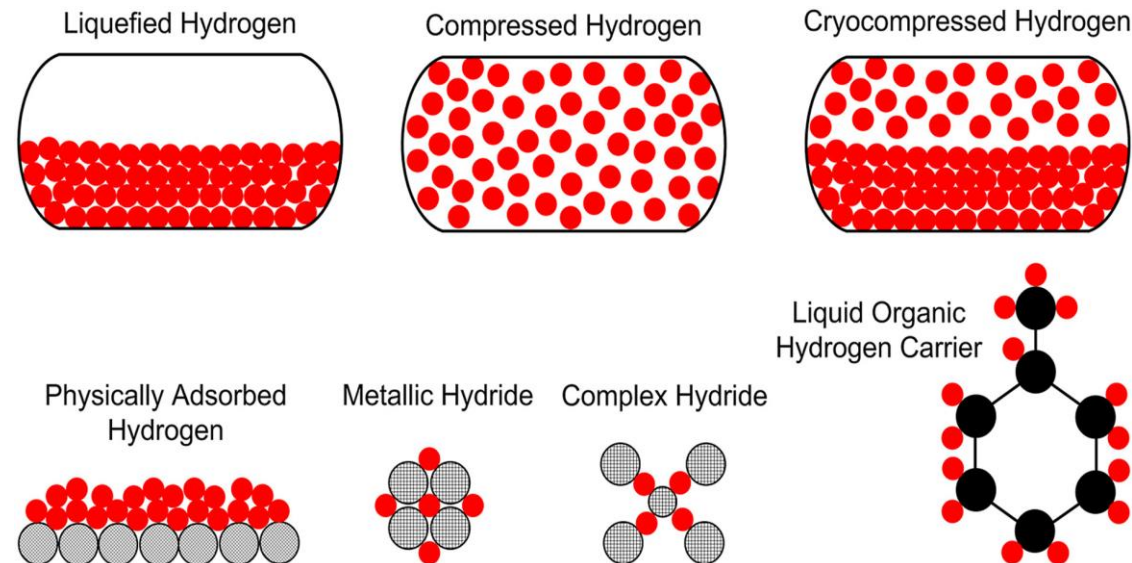
- Under R & D trials

Key challenges of the materials and their solutions

- All the materials used in storing hydrogen have high specific surface areas, but exhibit weak binding forces with Hydrogen (mostly by Vander Waals forces)
- A great deal of research show that metal decorations are effective to increase the bonding energy of hydrogen on sorption-based materials
- Hydrogen storage in carbon materials is increased by decorations of these materials with transition metal because of **hydrogen spillover phenomenon**
 - Hydrogen spillover is the migration/transfer of hydrogen atom from metal surface into the non-metal support or absorbent

- **Conclusion:**
- The Hydrogen storage technology is rapidly emerging as a fast alternative to fossil fuel but it needs further improvements in terms of infrastructure
- Two mechanism are involved in the hydrogen storage, physisorption and chemical absorption
- A wide variety of carbonaceous materials, metal organic frameworks hydrates etc are used as hydrogen storage materials
- The extensive and increasing research effort in hydrogen storage materials represents an exciting long-term research frontier

Physical and material-based methods of storage of Hydrogen





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HARISH N

9738889242

Department of Science and Humanities

ENGINEERING CHEMISTRY

Module VI- Electrochemical Sensors

Module Content :

- ***Classification of sensors***
- ***Types of Electrochemical Sensors***
- ***Electrochemical Sensors: Gas Sensor- O₂ Sensor***
Biosensors- Glucose Sensor

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Module VI- Electrochemical Sensors



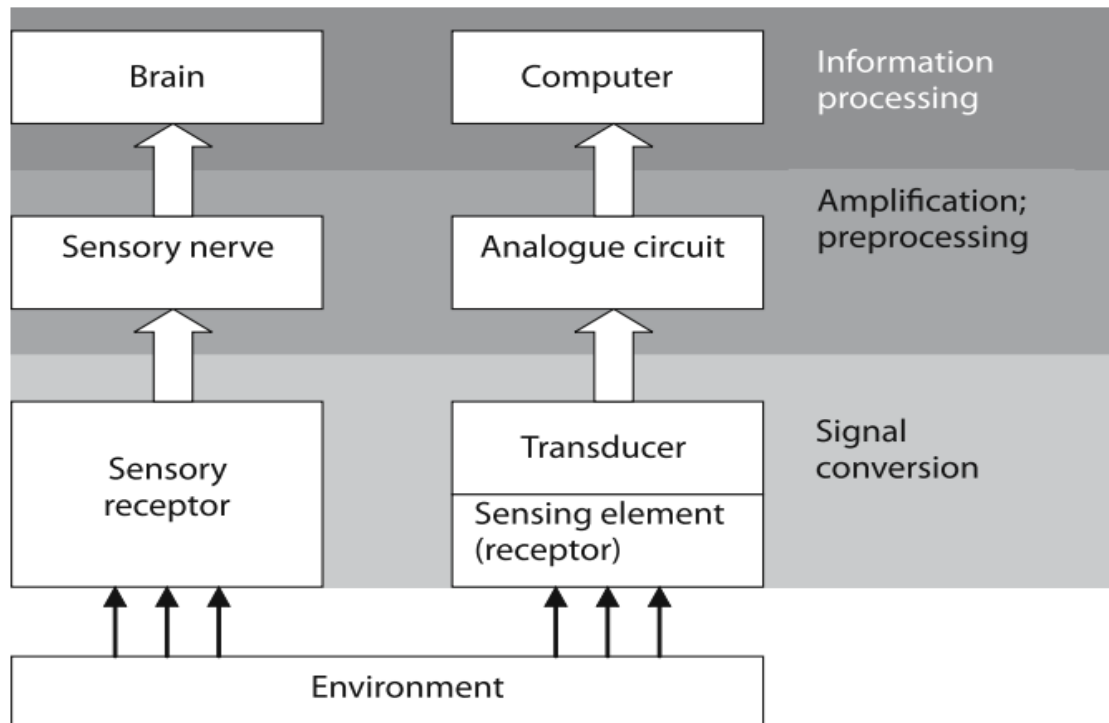
Class Content :

- *Classification of sensors*
- *Types of Electrochemical Sensors*

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Module VI- Electrochemical Sensors

- **SENSORS:** A sensor is a device that detects and responds to some type of input from the physical environment.
- The specific input could be light, heat, motion, moisture, pressure, or can be any molecule or substance such as analytes.



Signal processing in living organisms and in intelligent machines

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Module VI- Electrochemical Sensors

TYPES OF SENSORS

Sl No	Types of Sensors	Monitored property	Examples
1	Electrochemical Sensors a. Potentiometric Sensors b. Amperometric Sensors	a. Potentiometric -Voltage b. Amperometric - Current	Potentiometric - Ion selective electrode-pH sensor-Glass electrode/ Oxygen Sensor Amperometric - Biosensors- Glucose sensors (measures current)
2	Mass-Sensitive Sensors	Mass Sensing	Piezoelectric material –Quartz Crystal microbalance, piezoresistive- Silicon Carbide
3	Conductivity Sensors	Conductance (Resistance Sensing)	Platinum electrode – Probe with two electrode(Pt)
4	Capacitive Sensors	Capacitance (Potential Load Sensing)	Touchscreens- glass/polyester/polycarbonate/polyethylene

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Module VI- Electrochemical Sensors

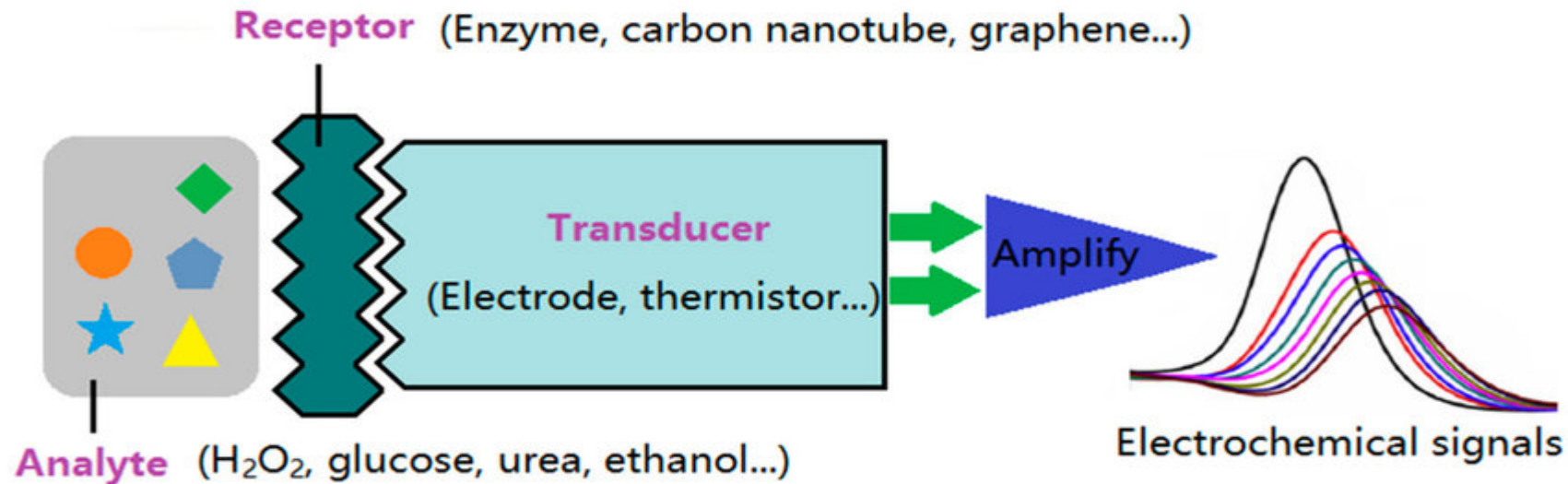
TYPES OF SENSORS

Sl No	Types of Sensors	Monitored property	Examples
5	Thermometric Sensors	Heat Sensing	Thermistors- Semiconductors (ceramics/polymers), Thermocouple-iron-constantan&copper-constantan Alloys Resistance Temperature Detectors (RTDs)- Platinum metal RTDs
6	Calorimetric Sensors	Enthalpy Change/ Heat transfer	Differential Scanning Calorimetry (DSC)- Thermocouples- metal/metal alloys (Pt-Pt/Rh) /(Ni-Cr/Ni-Al) Microcalorimeters- Aluminium, steel, Quartz.
7	Optical Sensors	Photon/Light	Photo detectors- photodiodes and phototransistors Optical fibre sensor- silica glass/Plastic Optical fibres (POFs), Gallium Arsinide, Ge-Si Solar Cells

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Module VI- Electrochemical Sensors

- **ELECTROCHEMICAL SENSORS:** A class of chemical sensors in which an electrode is used as a transducer element in the presence of an analyte.
- Electrochemical sensors transfers the effect of the electrochemical interaction of the analyte-electrode into a useful electric signal.



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Module VI- Electrochemical Sensors

Types of Electrochemical Sensors

- Potentiometric Sensors
- Amperometric Sensors

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Module VI- Electrochemical Sensors

TYPES OF ELECTROCHEMICAL SENSORS

1. POTENTIOMETRIC SENSORS

- Sensors that use the effect of concentration on the equilibrium of redox reactions occurring at the electrode-electrolyte interface of an electrochemical cell.
- The redox reaction takes place on the electrode surface:

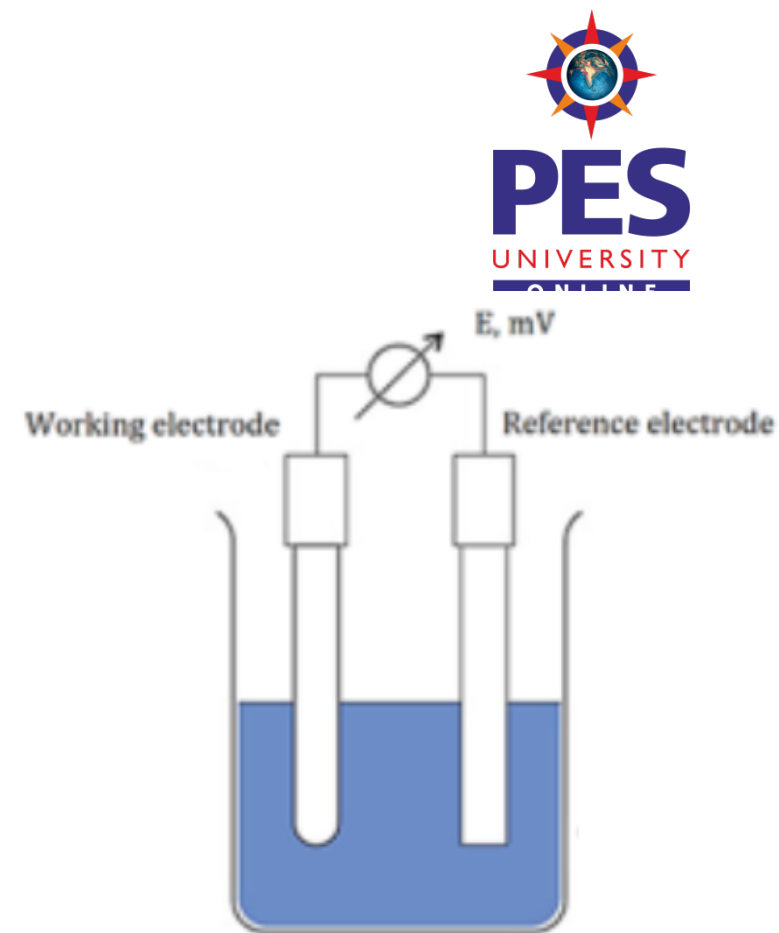


- Mainly determine the analyte concentration by measuring the variation of potential difference between working and reference electrodes at different analyte concentrations.

$$E_{\text{cell}} = E_o + \frac{RT}{nF} \log\left(\frac{C_o}{C_R}\right)$$

At 298K,

$$E_{\text{cell}} = \frac{0.0591}{n} \log\left(\frac{C_o}{C_R}\right)$$



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Module VI- Electrochemical Sensors



Examples of Potentiometric Sensors

Ion Selective Electrode Sensors

- Glass electrodes (e.g. pH-meter, Na^+ , K^+ , Li^+ Sensors)
- Solid membrane electrodes (e.g. based on AgX for X^- , and Ag_2S for other M^+).- Liquid membrane electrodes (e.g. containing a ligand for M^+ complexation e.g. Ca^{2+} and K^+ sensors).

Gas Potentiometric Sensors

- pH-meter-based gas detectors (e.g. CO_2 , NH_3 etc).
- Solid oxide sensors (e.g. zirconia-based O_2 sensor (λ -sensor))

Source: <http://photobiology.info/Visser-Rolinski.html>

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Module VI- Electrochemical Sensors



Potentiometric sensors

Advantages

- Applied to detect various analytes qualitatively and quantitatively.
- Easy to Construct and operate, more accuracy, sensitive and highly selective determination
- Smaller Volumes of analytes can be estimated
- Economically Viable

Limitations

- Requires calibration During estimation
- In Presence of other impurities can affect the potential Values
- By varying the temperature potential values also varies.

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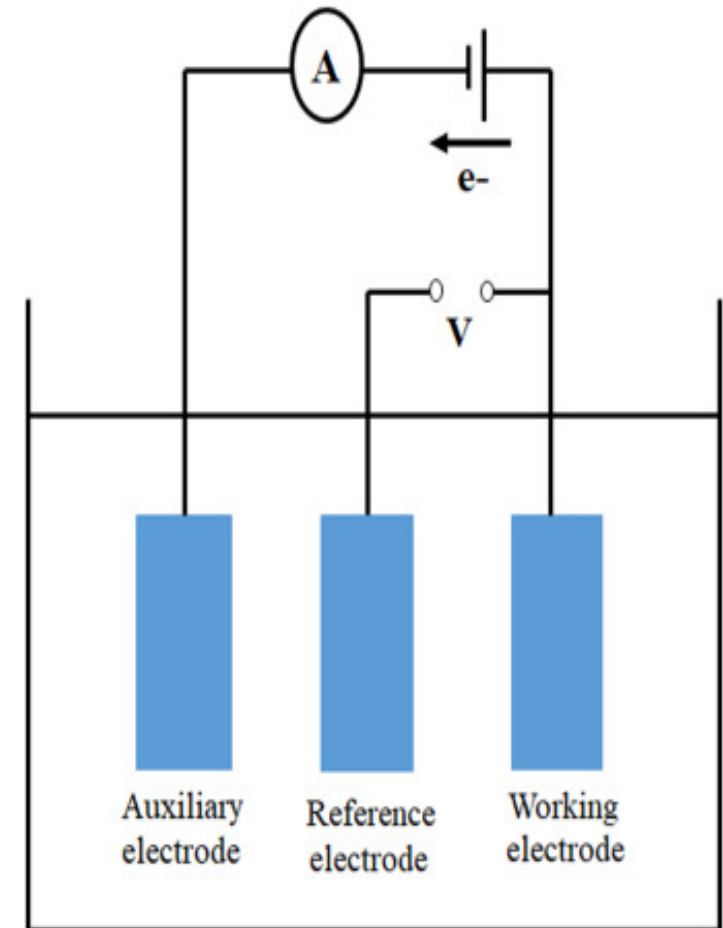
Module VI- Electrochemical Sensors

AMPEROMETRIC SENSORS

- Measure the current in response to detect the concentration of the analyte at a fixed potential.
- The applied potential drives the electron transfer reaction of the analytes, and the measured current indicates the analyte concentration.
- Amperometric sensors quantify the current output between Working and the reference electrode. The sensor is usually composed of 3 electrodes, that is working, auxiliary, and reference electrode



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Module VI- Electrochemical Sensors



Working of Amperometric Sensors

- The working electrode is the electrode at which the reaction of interest occurs, the reference electrode (e.g.; Ag/AgCl, Hg/Hg₂Cl₂) provides a stable potential compared to the working electrode.
- An inert conducting material (e.g.; platinum, graphite) is usually used as auxiliary electrode.
- A supporting electrolyte is used in controlled-potential experiments to eliminate electromigration effects, decrease the resistance of the solution and maintain the ionic strength constant.

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Module VI- Electrochemical Sensors

Advantages and limitations of amperometric sensors

ADVANTAGES	LIMITATIONS
• It can be used for estimation of reducible and non-reducible (Mg^{2+}, PO_4^{3-}, SO_4^{2-}) analytes	• Working potential applied should be for limited time or may lead to damage of the electrode
• More accurate and sensitive	• Coprecipitation gives inaccurate results
• Traces of reducible species can be determined accurately	• These sensors cannot be used with applied voltage more negative than -2V because hydrogen will be evolved
• Simple to operate and Easy to Construct	• Consumes more time to remove dissolved oxygen



THANK YOU

HARISH N

9738889242

Department of Science and Humanities



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Lata Pasupulety

Department of Science and Humanities

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Module 5- Energy storage devices – Oxygen sensor



Class content:

- *Oxygen sensor*
 - *Applications*
 - *Oxygen sensor in automobiles*
 - *Principle*
 - *Construction and working*

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Module 5- Energy Storage devices- Oxygen sensor



Oxygen sensor

An electronic device that **measures the proportion of O_2** in the gas or liquid being analysed

Applications of O_2 sensor:

- Divers use to measure partial pressure of O_2 in their breathing gas
- Scientists use as probes to measure respiration or production of O_2
- O_2 analyzers used in medical applications such as anesthesia monitors, respirators etc
- Used to measure exhaust gas concentrations of O_2 in IC engines

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Module 5- Energy Storage devices- Oxygen sensor



Role of oxygen sensor in automobiles

Principle :

- **Stoichiometric air/fuel ratio** ideal for combustion of gasoline is **14.7:1**
- Located in the **exhaust stream** and indirectly determine air to fuel ratio
- O_2 sensor allows engine control system to maintain ideal ratio across various engine operating systems
- It compares the amount of **O_2 in the exhaust** to the amount of **O_2 in the atmosphere**

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Module 5- Energy Storage devices- Oxygen sensor



- An output voltage is generated based on the **difference in O_2 concentration** in the exhaust and atmosphere
- An output voltage of **0.2V represents a lean mixture** and **0.8V represents a rich mixture**
- Ideal set point is around **0.45 V**
- The voltage is sent as feedback to Engine control unit which adjusts the air/fuel ratio back to stoichiometric value
- Lean mixture results in more of **NO_x emissions**
- Rich mixture leads to more of **CO , C particles and unburnt fuel emissions**
- Based on **solid state electrochemical fuel cell**
- Operates at a minimum temperature of **360°C**

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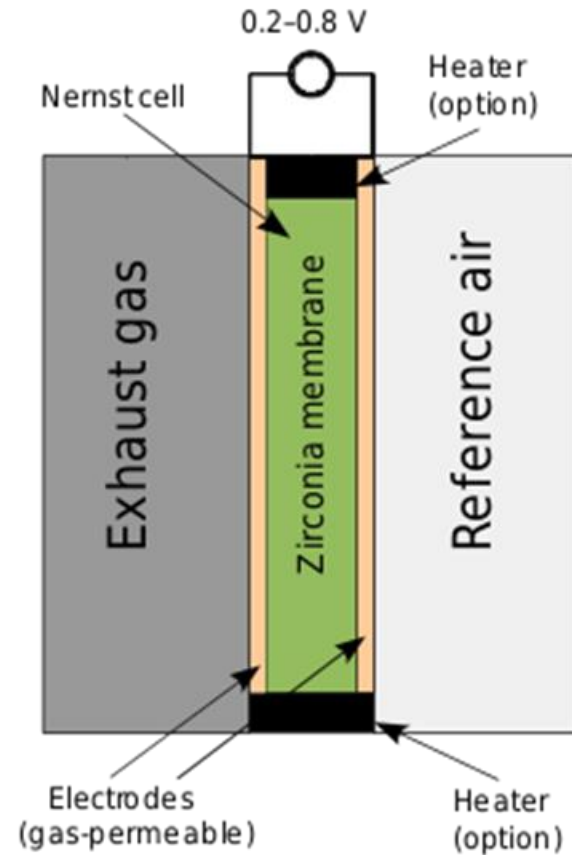
Module 5- Energy Storage devices- Oxygen sensor

Construction:

Anode : Pt

Cathode : Pt

Electrolyte : Zirconium oxide (ZrO_2)
doped with Yttrium oxide (Y_2O_3)

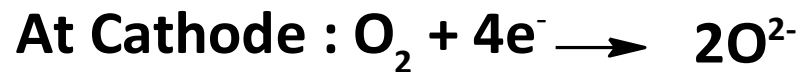
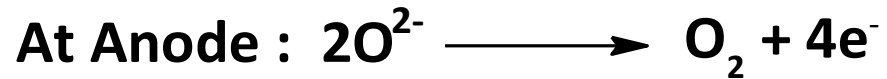


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Module 5- Energy Storage devices- Oxygen sensor



Working :



Cell voltage is given by

$$E_{\text{cell}} = \frac{2.303RT}{4F} \log \frac{P_1}{P_2}$$

where P_1 and P_2 are the partial pressures of O_2 in reference air and exhaust gas respectively

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Module 5- Energy Storage devices- Oxygen sensor



- Automobiles fitted with O_2 sensor must use **unleaded gasoline** as lead poisons the Pt catalyst and reduces its efficiency
- Symptoms of failure are – increased tail pipe emissions, increased fuel consumption and hesitation on acceleration



THANK YOU

Lata Pasupulety

Department of Science and Humanities

latapasupulety@pes.edu

+91 80 6666 3333 Extn 759



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HARISH N

9738889242

Department of Science and Humanities

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Module 6- Electrochemical Sensors



Class Content :

BIOSENSORS

GLUCOSE SENSOR

Components of glucose biosensor

Working of glucose biosensor

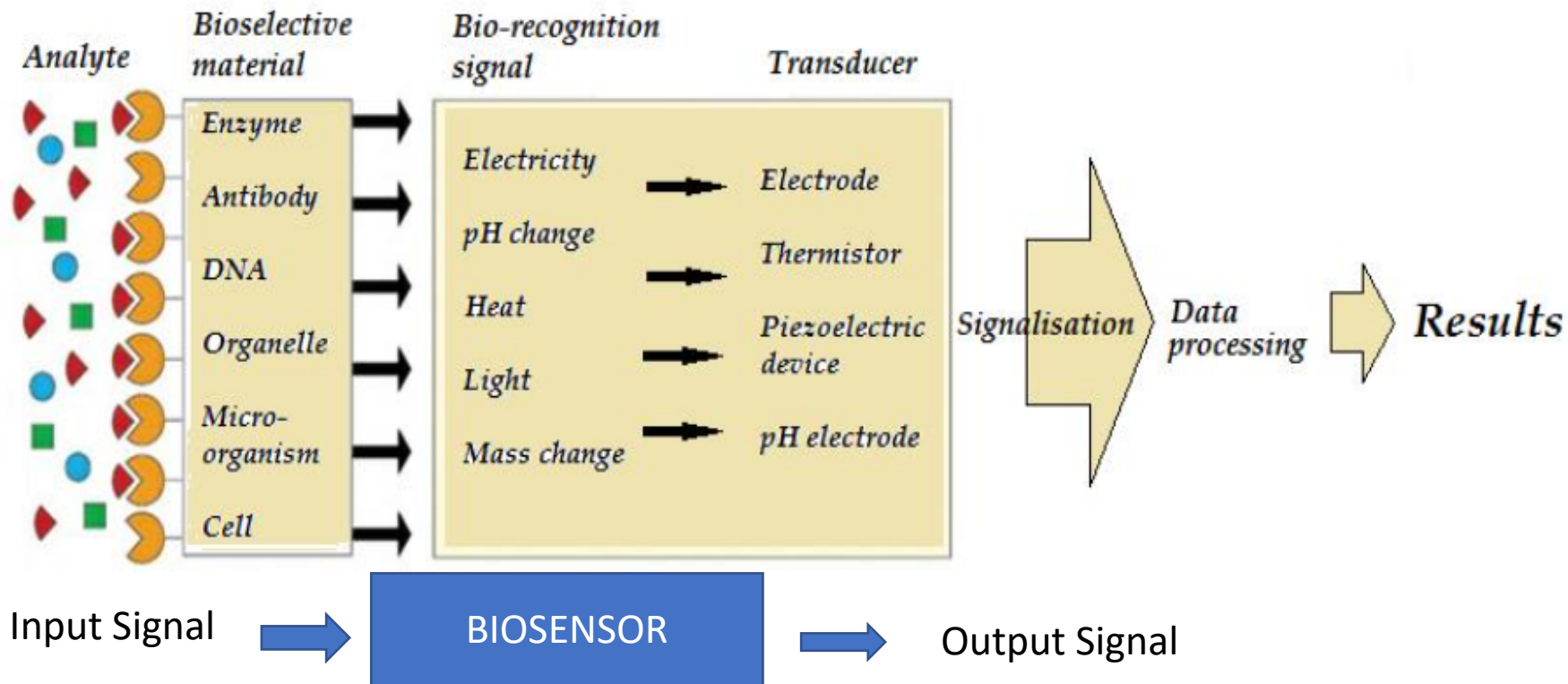
Limitations

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Module 6- Electrochemical Sensors

Biosensors

A biosensor is an analytical device which is used to determine the presence and concentration of a specific substance in a biological analyte.



Blood Glucose Monitoring

- Blood Glucose Monitoring is a way of checking the concentration of glucose in the blood using a glucometer.
- Provides quick response to tell if the sugar is high or low indicating a change in diet, exercise or insulin.
- Over time, it reveals individual of blood glucose changes.



Why Monitor Blood Glucose?

- Reduces risk of developing complications with diabetes.
- Allows diabetics to see if the insulin and other medications they are taking are working.
- Gives diabetics an idea as to how exercise and food affect their blood sugar.
- May prevent hypoglycemia or hyperglycemia



BLOOD SUGAR LEVEL CHART

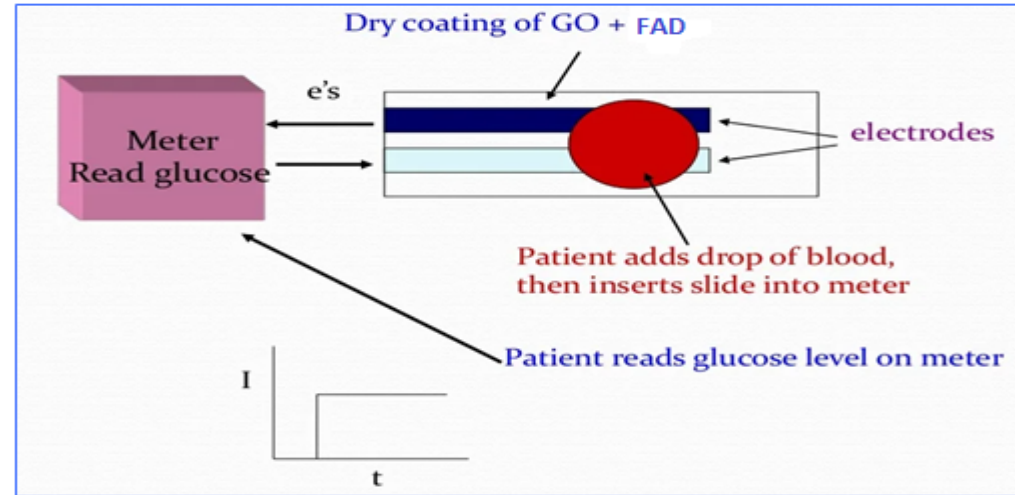
	FASTING	JUST ATE	3 HOURS AFTER EATING
NORMAL	80-100	170-200	120-140
PRE-DIABETIC	101-125	190-230	140-160
DIABETIC	126+	220-300	200+

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Module 6- Electrochemical Sensors

Components of Glucose Sensor

- Analyte
- Bioreceptor
- Transducer
- Electronics and display



Analyte

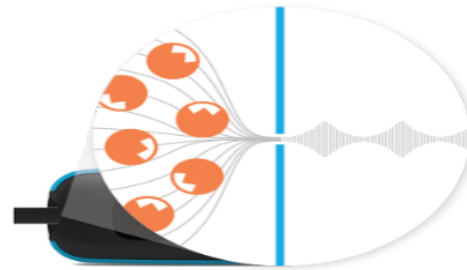


Bioreceptor



 = Glucose

Transducer



 = Enzyme

Display

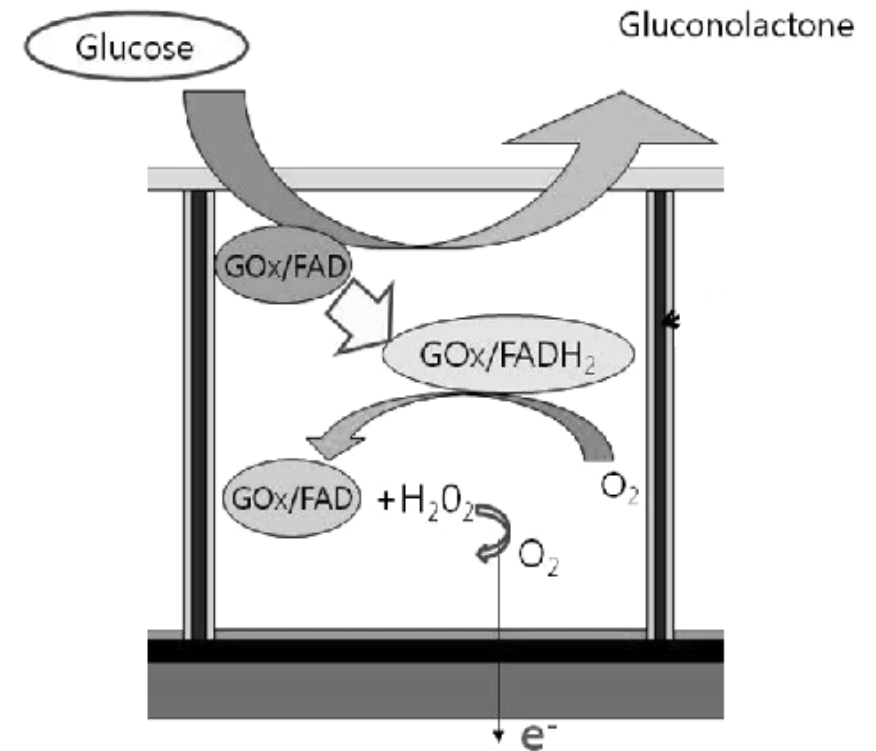
102 mg/dL

Working of Glucose Biosensor

- Developed by Updike and Hicks
- Enzyme Glucose oxidase(GOx) catalyze the oxidation of glucose by molecular oxygen producing glucolactone and hydrogen peroxide.
- In order to work as a catalyst, GOx requires a redox cofactor –flavin adenine dinucleotide (FAD), works as an initial electron acceptor and is reduced to FADH₂.
- $\text{Glucose} + \text{GOx} - \text{FAD}^+ \rightarrow \text{Glucolactone} + \text{GOx} - \text{FADH}_2$

Working of Glucose Biosensor

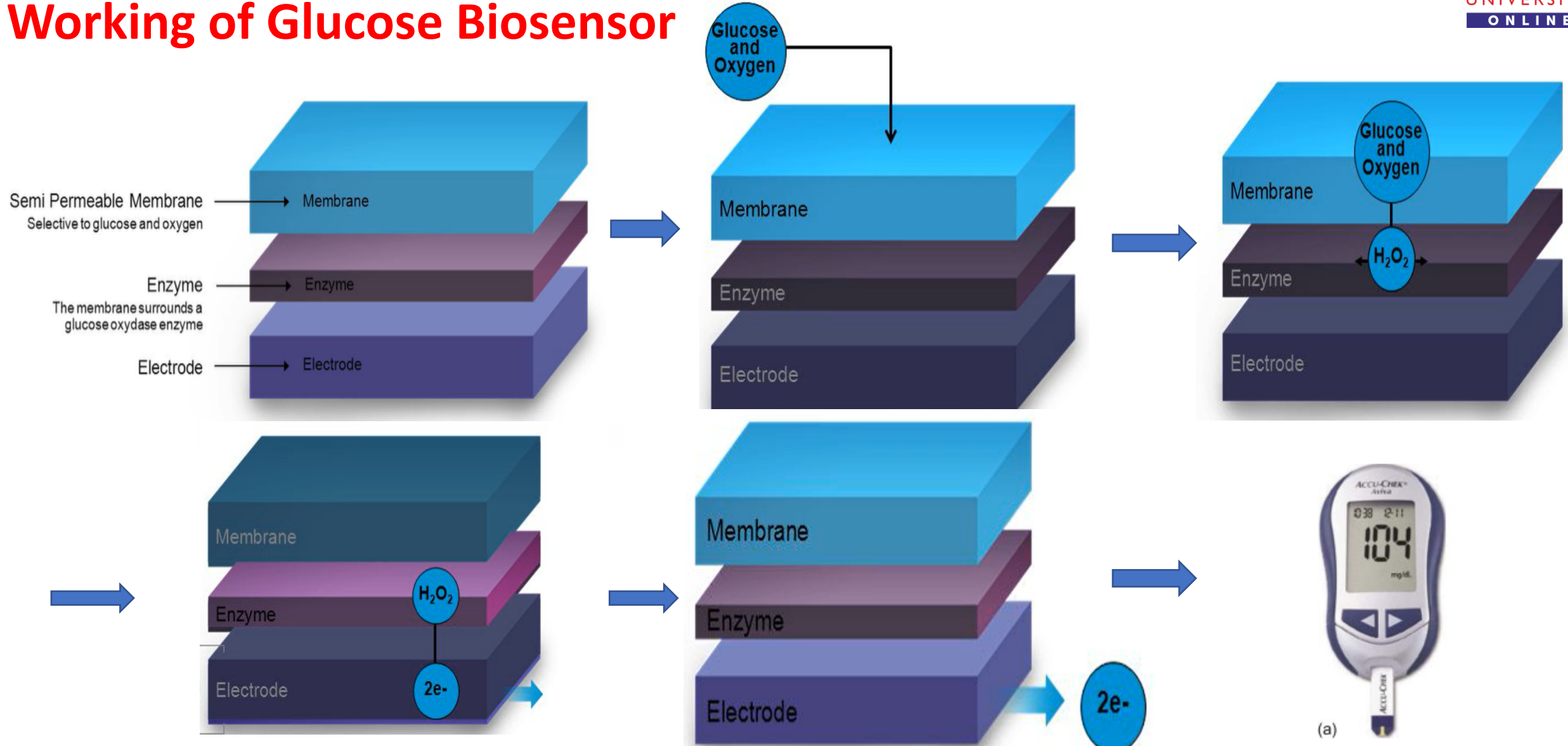
- The cofactor is regenerated by reacting with oxygen, leading to the formation of hydrogen peroxide
- $\text{GOx} - \text{FADH}_2 + \text{O}_2 \rightarrow \text{GOx} - \text{FAD} + \text{H}_2\text{O}_2$
- Hydrogen peroxide is oxidized at a platinum electrode.
- The number of electron transfers, at electrode surface is directly proportional to the number of glucose molecules present in the blood.
- $\text{H}_2\text{O}_2 \rightarrow 2\text{H}^+ + \text{O}_2 + 2\text{e}^-$



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Module 6- Electrochemical Sensors

Working of Glucose Biosensor



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Module VI- Electrochemical Sensors



Limitations of Glucose Sensors

- Extreme environmental conditions like haematocrit values, or medication interferences may potentially falsify blood glucose readings.
- Incorrect blood glucose readings may lead to treatment errors, for example, incorrect insulin dosing.



THANK YOU

HARISH N

9738889242

Department of Science and Humanities